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# Applications of quantitative laser sensors to kinetics, propulsion and practical energy systems

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**Abstract**

Laser diagnostic techniques play a large and growing role in combustion research and development. Here we highlight three areas where quantitative sensing based on laser absorption has had strong influence: chemical kinetics, propulsion, and practical energy systems. In the area of chemical kinetics, measurements in shock tubes of high-temperature reaction rate coefficients using species-specific laser absorption techniques have provided new and accurate answers to questions about combustion chemical processes. In the area of propulsion, wide-bandwidth measurements of flow temperatures, species concentrations, and velocity have provided engine designers with the necessary information to improve operation and performance. In the area of practical energy systems, real-time measurements of combustor operating conditions and emissions have enabled needed incremental improvements in large power plants and improved safety of operation. Yet, there is still more to be done, and opportunities for new applications will grow as laser sensors evolve. This review seeks to provide an overview of the current power and future potential of these modern diagnostic tools.
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*Keywords:* Laser absorption; Shock tubes; Propulsion; Combustors; Practical energy systems

## 1. Introduction

Over the past fifty years since the invention of the laser, diagnostic techniques utilizing laser light have played a growing role in combustion research and development. The importance of laser diagnostics to the field of combustion is evidenced by inclusion of diagnostics as one of the regular colloquium topics at the biennial international symposia on combustion and by the number of invited presentations on this topic at these meetings. As prominent examples, laser diagnostics in combustion was the subject of Wolfrum's Hottel Lecture at the 27th Combustion Symposium in 1998 as well as one of the invited 50th Anniversary Lectures at the 30th Combustion Symposium in 2004 [1,2].

While there have been numerous reviews of laser diagnostics in combustion, many of these have focused on new measurement concepts and the associated spectroscopy, or on application of laser diagnostics to traditional combustion flowfields such as flames or devices such as internal combustion engines, where the results were often qualitative. In fact, it has proven difficult to render many of the most powerful laser diagnostics highly quantitative, for various reasons. Of the candidate laser diagnostic methods currently in use for species-specific sensing, the most quantitative method is laser absorption, a line-of-sight technique, and in some cases laser-induced fluorescence, a variant of absorption that can yield spatially resolved information.

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The present paper seeks to demonstrate the progress made in combustion science as a result of quantitative laser-based sensing, with emphasis on absorption methods and, in one case, planar laser-induced fluorescence. To provide useful depth, the paper will focus on three application domains of combustion, namely combustion kinetics, propulsion, and combustion in practical energy-conversion systems (excluding internal combustion engines). Particular objectives of the paper are to illustrate the growing impact of quantitative laser diagnostic methods on combustion science and to communicate both the power and relative simplicity of these methods to those with interests and expertise outside of spectroscopy.

## 2. Chemical kinetics

Chemical kinetics has long been a fundamental discipline within the combustion field, and indeed this topic typically attracts a significant population of papers and attendees at the biennial international combustion symposia. The subject attracts experimentalists, theorists and computational scientists, with important collective advances being made in the field as a result of the synergism across these domains. The most common experimental systems used for fundamental studies of combustion kinetics are shock tubes, rapid compression machines, flames of different types, and flow reactors of different design including well-stirred reactors and both laminar and turbulent flow reactors. In order to provide sufficient depth, we will focus here on shock tube studies, recognizing that several excellent papers on and reviews of laser applications to kinetics in other systems are available elsewhere [3,4].

The laser diagnostic method with greatest current use and impact in shock tube kinetics is narrow-linewidth absorption with a continuous-wave (CW) source, often tunable in wavelength. Applications have utilized the full spectral range from the ultraviolet to the mid-infrared. Particular progress has been made in recent years owing to increasing availability of user-friendly laser systems of reasonable cost.

In the following sub-section, we will first summarize the relevant physics and detection capability of laser absorption systems and then give a brief overview of shock tube methods for studies of gas-phase combustion kinetics. In subsequent sub-sections, we present examples of studies aimed at different aspects of combustion kinetics, including: elementary reactions, pressure-dependent reactions, reactions of combustion-generated pollutants, fuel pyrolysis reactions, and fuel oxidation reactions. Following these sub-sections, we summarize some recent advances in shock tube operation and modeling, both of which have led to more accurate kinetics results, and close with a sub-section pointing out future needs and opportunities in laser absorption/shock tube kinetics studies.

In this paper, we will focus on gas-phase reactions. Excellent pioneering work on laser absorption studies of heterogeneous reactions in shock tubes has been reported elsewhere, notably by Roth and co-workers [5,6].

### 2.1. Diagnostics fundamentals and facilities

The combination of shock tubes and laser absorption provides a powerful test bed for chemical kinetics studies at high temperatures. Shock tubes generate near-constant-volume conditions with an effectively instantaneous step change to well-known initial temperatures and pressures. CW laser absorption can provide highly sensitive and quantitative measurements of species time-histories. With good experimental design, this combination of well-known reaction conditions and accurate species time-histories can yield unique kinetics data without significant influence from other processes. Here we describe the fundamentals of laser absorption and shock tube operation.

#### 2.1.1. Absorption theory and lineshapes

Absorption spectra can be divided into two categories: showing either discrete features or broad features. The absorption theory of both categories is essentially the same, but slightly different equations are used.

For discrete absorption features, a spectrally narrower laser probe beam is needed to take advantage of the sensitivity provided by the maximum peak absorption. CW laser radiation can provide this narrower spectral bandwidth and permits a simple theoretical description of line-of-sight absorption for the uniform concentration profile generally found in shock tubes. The fractional transmission of a monochromatic incident beam obeys the Beer–Lambert law:

$$T_\lambda = (I/I_0)_\lambda = \exp(-k_\lambda P_{\mathrm{Total}} X_i L) \quad (1)$$

where $P$ is the total pressure (atm), $X_i$ is the mole fraction of the absorbing species $i$, $L$ is the shock tube diameter (cm), and $k_\lambda$ is the absorption coefficient ($\mathrm{atm}^{-1}\,\mathrm{cm}^{-1}$) at wavelength $\lambda$ [7]. The absorption coefficient $k_\lambda$ is the product of the integrated line strength S ($\mathrm{atm}^{-1}\,\mathrm{cm}^{-2}$) and the lineshape function $\varphi(\nu - \nu_0)$ (cm); the former is a function of temperature and the specific transition, and the latter is a function of pressure, temperature, composition and diagnostic wavelength. The integrated line strength is generally known from existing databases or can be derived from calibration experiments that use known concentrations of absorbers and measured or known lineshapes. [8].

For most shock tube/laser absorption experiments, a Voigt lineshape profile is employed. A Voigt profile represents the convolution of a Gaussian lineshape function resulting from Doppler broadening with a Lorentzian lineshape function resulting from collisional broadening [9,10]. Doppler broadening is a simple function of the temperature, but the coefficient for collision broadening must be measured in a heated cell or the shock tube itself [11,12]. Collision-induced shift in line position must also be accounted for at elevated pressures.

A critical first step in developing an absorption diagnostic is to perform spectral simulations that identify promising absorption transitions for the target species. As an example, spectral absorption data for the $R_1(5)$ OH $A-X(0,0)$ line at 306.69 nm used for OH concentration measurements in numerous shock tube kinetics studies are shown in Figs. 1 and 2 [13,14]. This spectral line is well separated from adjacent lines allowing for a simple description of its absorption lineshape, and its large line strength provides strong absorption and thus high sensitivity to OH concentration. The CW ring dye laser radiation used to probe this transition is effectively monochromatic, with a line width that is significantly smaller than the OH absorption line width. Knowledge of collisional width and shift for this transition are clearly important as indicated in Fig. 2; at 1500 K and 4 atm, the collisional shift is 0.05 $cm^{-1}$, the Doppler width is 0.22 $cm^{-1}$, and the total linewidth is ~0.32 $cm^{-1}$ [15].

A survey of species measured, wavelengths and laser systems used in shock tube experiments up to 2001 is given by Davidson and Hanson [16]. Since then, the list of species and wavelengths in use has increased substantially, particularly in the infrared where new commercially available diode lasers and existing gas lasers have enabled sensitive detection of $H_2O$, $CO_2$, $CH_4$ and higher alkanes, and other hydrocarbons [17–21].

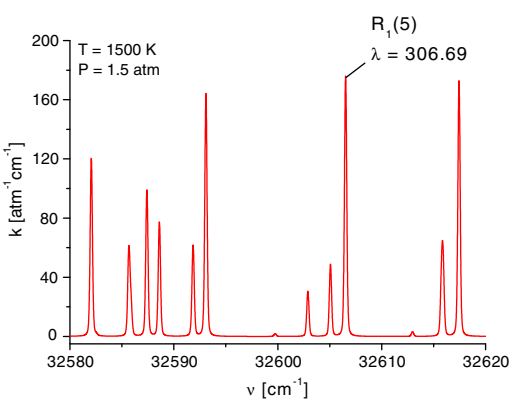


Fig. 1. Details of the OH $A-X(0,0)$ $R_1(5)$ absorption feature near 306.7 nm: absorption coefficient versus wavenumber.

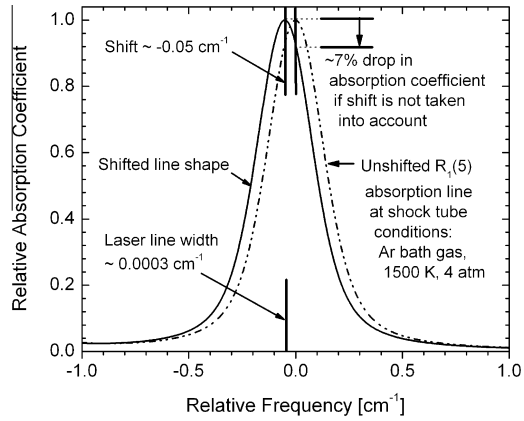


Fig. 2. Spectral details of the $R_1(5)$ lineshape.

A schematic of a typical experimental setup for a ring dye laser absorption system to measure OH at 306.7 nm is given in Fig. 3. Radiation at the fundamental wavelength, 613.4 nm, is generated in a ring dye laser cavity, which is pumped by a solid-state CW, frequency-doubled YAG laser (532 nm). Mode quality is monitored using a scanning interferometer and wavelength is monitored using a wavemeter. Frequency-doubled radiation is generated intracavity using an AD*A crystal. The output radiation is split into a reference beam ($I_0$) which carries information about laser amplitude variations, and a probe beam ($I$) which is sent through a train of optics to focus it through the test gas in the shock tube where it is absorbed by the target species; both beams are collected on fast photodetectors. The reference beam is used to remove common-mode noise and produce a fractional transmission ($I/I_0$) signal which can be directly related to species concentration using the Beer–Lambert law [22].

Fig. 4 shows representative detection limits using laser absorption for a variety of different species. These values are for particular transitions commonly used, though they are not necessarily the strongest absorbing feature for a particular band or species. The minimum detectivities are based on signal-to-noise (SNR) = 1 for bandwidths of 1 MHz, pathlengths of 10 cm, with a minimum detectable absorption of 0.1%. (Application of wavelength-modulation spectroscopy techniques with harmonic detection, i.e. WMS-2f, discussed later in this paper, can improve detection limits by at least a factor of 10.).

### 2.1.2. Shock tubes

Shock tubes can provide near-constant-volume reaction conditions with well-defined temperatures and pressures for kinetics investigations. Using these facilities, temperatures over the range of 500–5000 K and higher, and pressures from sub-atmospheric to 500 atm and higher, can be

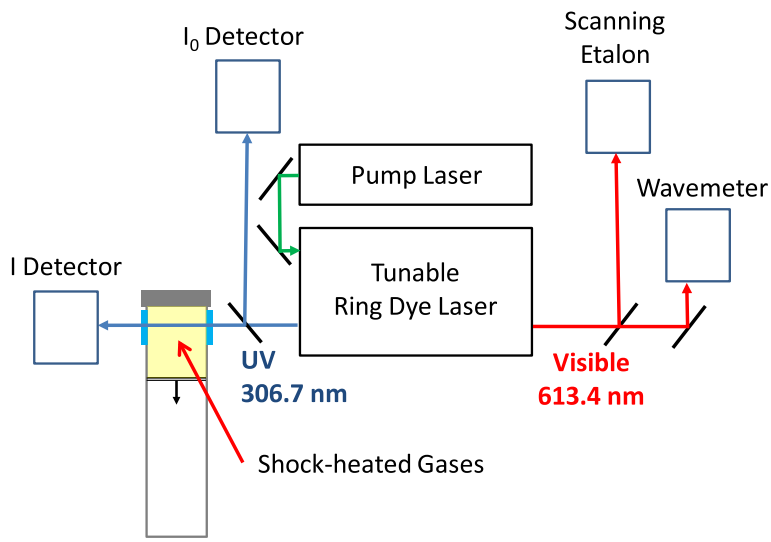


Fig. 3. Representative shock tube/OH laser absorption setup.

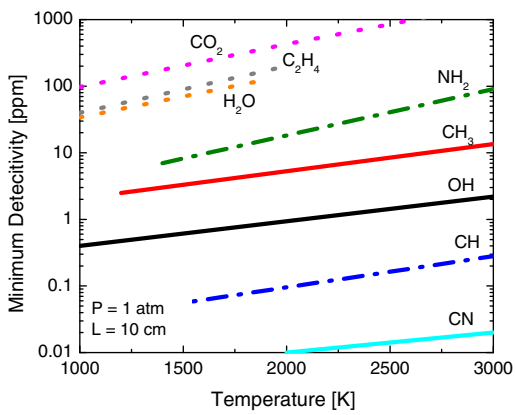


Fig. 4. Minimum detectivity limits for shock tube/laser-absorption measurements. Dotted lines: IR transitions; dot-dashed lines, visible transitions; solid lines, UV transitions.

achieved. The test gas volume, typically the stagnated region behind the reflected shock wave, experiences near-instantaneous heating with a spatially uniform mixture distribution. Transport phenomena and heat transfer losses to boundary surfaces can generally be neglected, particularly in shock tubes with large diameters, making them nearly ideal reactors for kinetic studies.

A shock tube is simply a long tube, closed at both ends and divided into two sections, usually by a diaphragm [23]. High-pressure gas in one section of the tube, the driver section, bursts the diaphragm and generates an incident shock wave that propagates through the low-pressure section of the tube, the driven section. The wave structure in the shock tube during an experiment is summarized in the $X - t$ diagram in Fig. 5. Behind the incident shock wave, the test gas is compressed, heated and swept forward. The conditions immediately behind the incident shock wave are described by the normal shock equations [24]. This incident shock wave reflects from the closed end wall of the shock tube and propagates back along the driven section. Conditions immediately behind an ideal reflected shock wave are also well-described by normal shock equations, and are used to achieve the desired test temperature and pressure. Of note is that the test gas behind the reflected shock wave is stagnated, and if there is only small energy change associated with reactions, the test gas near the end wall behaves nearly as a constant volume reactor and hence is readily modeled.

Low-pressure kinetics studies are best performed in a shock tube of large diameter. Such shock tubes have small, or insignificant, boundary layer effects, and shock wave flows in these tubes more closely approximate ideal behavior. The following is a description of a high-purity shock tube facility used for chemical kinetics studies at Stanford. The shock tube has a 6 inch (15.24 cm) I.D. and was fabricated from electro-polished 304 stainless steel. The driven section has a length/diameter ($L/D$) ratio of 70 and uses a helium-filled driver section with an $L/D$ of 30 to burst polycarbonate diaphragms (against a crossed-knife arrangement in a short square section of the shock tube). The shock tube is sealed with Viton o-rings, is usually operated at room temperature (though it can be heated to 100 °C), and can be turbo-pumped to an ultimate pressure below $10^{-6}$ torr and to a leak and outgassing rate of better than $3 \times 10^{-6}$ torr/min. Typically, reflected shock wave experiments are

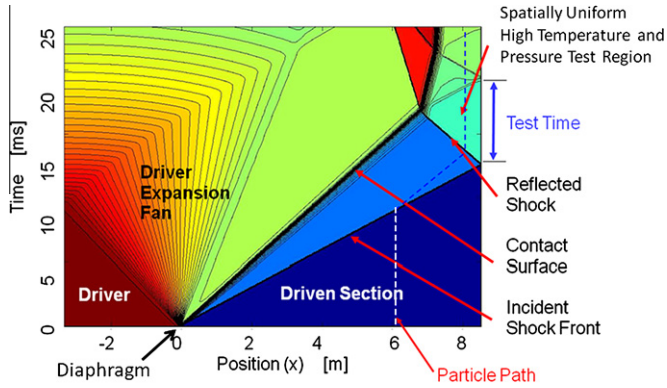


Fig. 5. Shock tube $X - t$ diagram [25]. Particle path shown in blue dashes.

performed hourly, and over this time frame the ultimate shock tube pressure before filling is about $5 \times 10^{-6}$ torr with a leak and outgassing rate of $10^{-5}$ torr/min.

Critical to the use of shock tubes as reaction vessels is the accurate determination of temperature. Both the incident and reflected shock pressures and temperatures are calculated using the incident shock Mach number. Incident shock speeds are normally determined by measuring the incident shock arrival times using a series of piezo-electric pressure transducers (PZT) spaced at intervals over the last several meters of the driven section. Current accuracies of the incident shock speed measurements can provide reflected shock initial temperature and pressure uncertainties of ±0.5% or better.

A schematic of a laser-instrumented shock tube setup at Stanford is shown in Fig. 6. Diagnostic measurements are made close to the end wall, typically 1–2 cm away.

### 2.2. Elementary reactions

Combustion chemistry is typically modeled with detailed mechanisms comprised of many reactions. This requires specification of rate coefficients which are derived from various sources: theory, experiments on single (elementary) reactions, and experiments with many reactions contributing. Where possible, direct study of individual reactions is best, and we begin here with examples of this approach.

The capability in shock tube experiments to accurately determine the temperature, pressure and concentration of initial reactants, and the ability to follow time-histories of these reactants or transient intermediates and products using laser absorption, allows the kinetics researcher to design experiments that can enable the direct or near-direct determination of individual reaction rates. These experiments can be modeled using current chemical mechanism solver codes such as CHEMKIN or CANTERA [26,27]. It is good practice also to use sensitivity analysis, e.g. using SENKIN, to plan optimized experiments, seeking conditions that take best advantage of the diagnostic sensitivity and signal-to-noise ratio available [28].

Ignition processes are sensitive to many different reactions. However, of these reactions the chain-branching reaction $H + O_2 \rightarrow OH + O$ (1) plays a pivotal role in all hydrocarbon oxidative chemistry. This reaction also provides an excellent example of the effectiveness of shock tube/laser absorption experimentation in determining elementary reaction rate coefficients.

#### 2.2.1. Example: $H \pm O_2 \rightarrow OH \pm O$

Accurate values of $k_1$, the rate coefficient of Reaction (1), are required to enable accurate predictions of oxidative processes in hydrocarbon fuels. Hence, this reaction has been the subject of a wide array of studies [29–33]. In many cases, such as the prediction of ignition delay time, and formation rates of intermediate species, small changes in the value of $k_1$ can have a large effect on mechanism performance. At high temperatures (1450–3370 K), measurements using OH laser absorption provide the most reliable values for the rate coefficient of this reaction [34]. At lower temperatures, somewhat larger uncertainty exists, in part because of reliance on earlier, slightly more scattered H-ARAS measurements [35].

An example of sensitivity analysis used in planning recent shock tube experiments at Stanford is shown in Fig. 7, indicating that in the lower temperature range, 1100–1500 K, accurate measurements of the water vapor time-history in rich $H_2/O_2$/Argon mixtures would allow near-direct access to Reaction (1). At these conditions, $H_2O$ concentration is primarily sensitive to $k_1$, with only small sensitivity to the rate coefficients of the two reactions $H + H_2O$ and $O + H_2$.

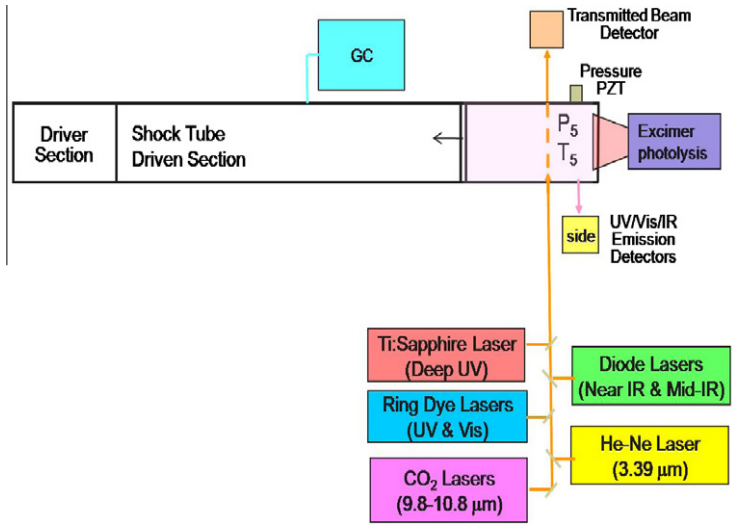


Fig. 6. Schematic of shock tube/laser absorption setup at Stanford.

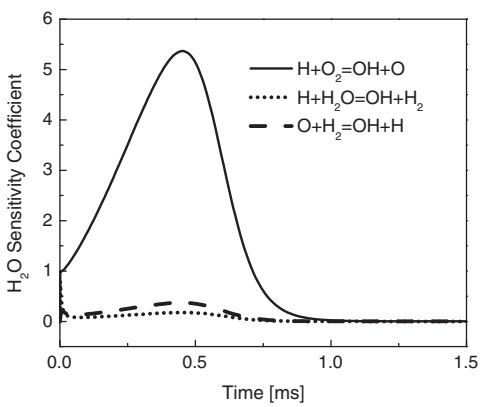


Fig. 7. $H_2O$ sensitivity plot used in experimental planning. Reflected shock conditions: 0.1% $O_2$, 0.9% $H_2$, 99% Ar, 1472 K, 1.83 atm.

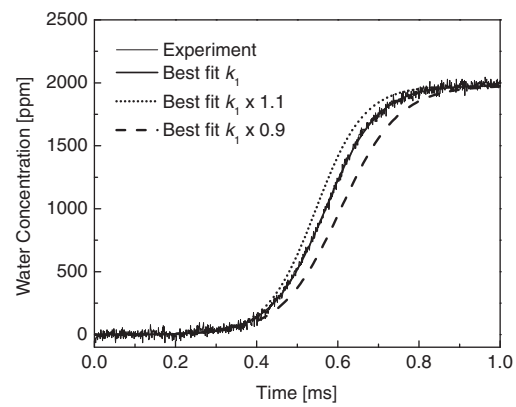


Fig. 8. Typical laser absorption history for $H_2O$ in a reflected-shock experiment [37]. Reflected-shock conditions: 0.1% $O_2$, 0.9% $H_2$, 99% Ar, 1472 K, 1.83 atm.

Water concentration was measured using IR DFB (infrared distributed feedback) laser absorption at 2550.96 nm (3920.09 $cm^{-1}$) within the $\nu_3$ fundamental vibrational band. Spectral parameters for this transition are available from the HITRAN database, [36] and the collisional-broadening coefficient (half-width at half maximum per atm) for argon (not available in HITRAN) was measured in separate shock tube experiments.

Representative data from this study are shown in Fig. 8. The high SNR of these measurements allows for very sensitive fitting of the water profile. Modeling with $k_1$ variations of 10% are shown in Fig. 8. Further details of this study can be found in Hong et al. [37].

The best-fit rate coefficients $k_1$ for these experiments are presented on the Arrhenius plot in Fig. 9. Substantially reduced scatter is evident in these measurements relative to the earlier H-ARAS measurements of Pirraglia et al. [35]. The new low-temperature $H_2O$-based data are in excellent agreement with the higher temperature OH-absorption-based measurements of Masten et al. obtained in our laboratory 20 years ago [34].

### 2.2.2. Excimer laser photolysis

The design of shock tube/laser absorption experiments to study elementary reaction of transient radicals is limited by the availability of suitable precursors for these species. Reactions of $CH_3$, for example, can be studied using azomethane, a stable compound at room temperature that readily decomposes to form two $CH_3$ radicals and $N_2$ at high (>1000 K) temperatures. Similarly, atomic O can be produced from decomposition of

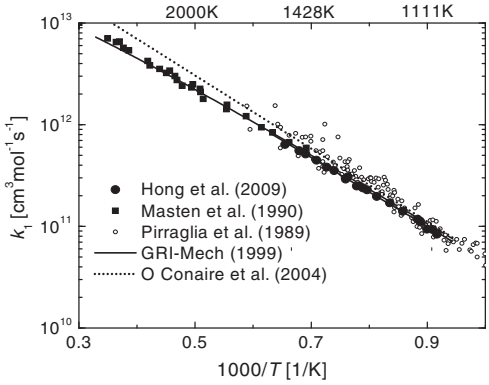


Fig. 9. Arrhenius plot for $H + O_2 \rightarrow OH + O$. Previous studies and modeling are found in Refs. [34,35,37–39].

$N_2O$ at temperatures greater than about 1700 K, H may be obtained from decomposition of $C_2H_5I$ above 1200 K, and OH can be produced from tert-butyl hydroperoxide at temperatures above 1000 K; but all such methods are generally constrained in their temperature range and may suffer from not being instantaneous.

A more versatile (but experimentally more complex) approach to producing radicals and atoms is to use laser photolysis. Commercially available excimer lasers can produce high-energy pulses of relatively narrow-linewidth radiation at selected wavelengths from 157 to 308 nm. This radiation can be used to instantaneously dissociate stable reactants to form transient intermediate species for elementary reaction rate determinations. For example, N- and O-atoms can be formed by 193 nm ($ArF_2$) excimer photolysis of NO [40]. These radicals can then react with their precursor (NO) as well as with other reactants added to the initial test gas mixture. Accurate measurements of the rate coefficient for $N + NO \rightarrow N_2 + O$ were made by measuring the decay of N-atoms formed in this fashion using ARAS (atomic resonance absorption spectroscopy) [40].

As another example, low-temperature measurements of fuel–nitrogen reactions, in particular reactions of HCN, have been difficult to access with conventional pyrolysis-based experiments. However, excimer laser photolysis experiments using ArF radiation (193 nm) to form CN radicals from $C_2N_2$ allowed direct measurement of the reaction rates of $CN + H_2$ and $CN + HCN$ [41]. High-accuracy measurements of these reaction rates at low temperatures (940–1840 K) provided needed constraints on theoretical expressions for these reaction rates not available from room-temperature and higher-temperature measurements alone.

### 2.3. Pressure-dependent reactions

Shock tubes can provide near-constant-volume reactions conditions for a wide range of pressures and this capability can be used to study pressure-dependent reactions. A good example is the study of ethane decomposition by Oehlschlaeger et al. who used laser-absorption measurements of the product species $CH_3$ [42].

#### 2.3.1. Example: ethane decomposition

The rate coefficient of the ethane decomposition reaction, $C_2H_6 \rightarrow CH_3 + CH_3$, was measured in the fall-off regime at high temperatures in a shock tube using narrow-line laser absorption of $CH_3$ at 216.6 nm. These measurements covered the temperature range 1343–2034 K and the pressure range 0.13–8.4 atm. The decomposition rate coefficient was determined by monitoring the formation of $CH_3$ immediately behind shock waves and modeling this formation with a detailed kinetic model. The pressure dependence of this reaction can be characterized by the pressure fall-off model of Troe [43], as well as by RRKM/master equation simulations [44].

Radiation at 216 nm was generated using a ring dye laser combined with frequency-doubling (either intra- or extra-cavity doubling with β-BBO crystals). The $CH_3$ $B(^2A_1') - X(^2A_2')$ transition at 216 nm is composed of two prominent features, the P + Q and R band peaks, and provides very strong absorption even for low concentrations of $CH_3$.

The decomposition of ethane has been studied extensively [45–54]. While high-quality data exist at lower (296–906 K) temperatures [54], theoretical predictions of the high-temperature, high-pressure rate coefficients do not agree with each other or with available experimental data. High-temperature data with very low scatter were thus needed to resolve these differences.

Direct measurement of $CH_3$ time-histories by laser absorption provides high sensitivity to the ethane decomposition reaction. Representative $CH_3$ data from Oehlschlaeger et al. and the sensitivity analysis justifying this approach are shown in Fig. 10 [42]. $CH_3$ shows insignificant sensitivities to two other possible reactions: $CH_3 + C_2H_6$ and $CH_3 + CH_3 \rightarrow C_2H_5 + H$. This highly sensitive direct measurement is possible because of the very low initial reactant concentrations (200 ppm ethane) that can be employed due to the large $CH_3$ absorption signal possible at 216 nm.

Because of the low scatter in these data, as seen in Fig. 11, small but significant variations in the ethane decomposition rate with pressure can be measured. Agreement with RRKM predictions was found to be good [42,44].

### 2.4. Pyrolysis systems

The hierarchical extension of hydrocarbon reaction mechanisms to progressively larger reactants is based on the addition of sub-mechanisms that convert these larger reactants to smaller molecules

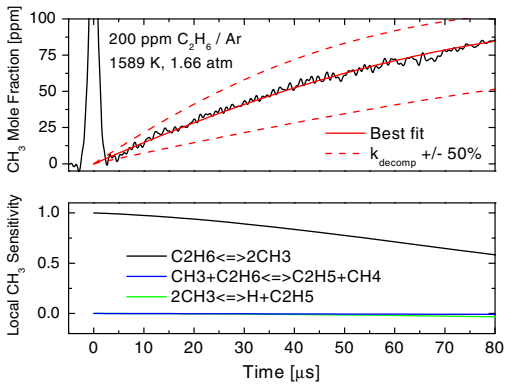


Fig. 10. $CH_3$ concentration time-history during ethane decomposition and model sensitivity. Adapted from Ref. [42].

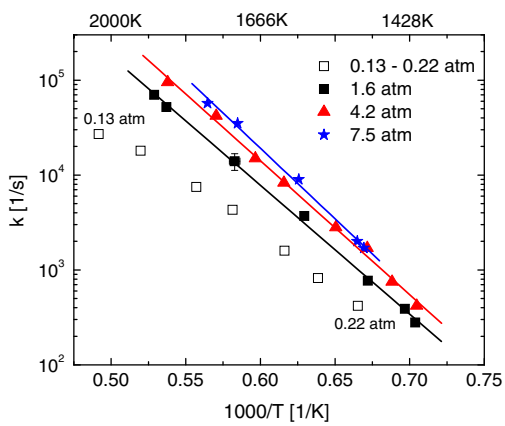


Fig. 11. Unimolecular ethane decomposition rate: Arrhenius plot. Solid lines are experimental best fit to data. Adapted from Ref. [42].

already incorporated in the base mechanism. Typically, there are multiple paths for this initial decomposition. Figure 12 provides an example of the initial primary pathways for *n*-heptane pyrolysis illustrating some of intermediate species needed for a complete model. In the case of *n*-alkanes, the initial dominant pathways for fuel decomposition during pyrolysis or oxidation are nearly the same. When an *n*-alkane is exposed to a high-temperature environment, it first decomposes via C–C bond scission, and then is attacked by the growing radical pool. The large radical intermediates that are formed undergo further cracking, primarily from beta scission, and producing small molecular fragments (H, $CH_3$, $C_2H_5$, $C_2H_4$, $C_3H_6$, etc.). Measurements of the concentration time-histories for these intermediate species place strong constraints on theoretically estimated values for these rates and pathways. Shock tube/laser absorption methods can provide this type of data; in fact, the study of

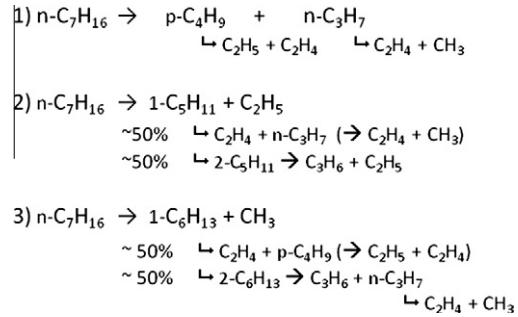


Fig. 12. Schematic of initial pathways in *n*-heptane decomposition during high-temperature pyrolysis. Courtesy of Dr. David Davidson, Stanford University (2010).

the initial decomposition steps of fuel decomposition is particularly suited to shock tube experiments because of the simpler chemistry that occurs at early time during pyrolysis.

### 2.4.1. Example: n-heptane decomposition

Decomposition rates can be measured either by following the reactant or the product time-history. In the case of the higher normal alkane, *n*-heptane, monitoring the reactant, *n*-heptane, and one of the products, $CH_3$, in separate experiments has been used to give a separate check on the experimental validity of the measurement of the overall rate coefficient of the reaction *n*-heptane → products.

In one *n*-heptane study, methyl radical concentration time-histories were measured using deep-UV laser absorption at 216 nm during the pyrolysis of *n*-heptane behind reflected shock waves over temperatures of 1100–1560 K and pressures of 1.6–2.0 atm [55]. In the second study *n*-heptane concentration time-histories were measured using infrared DFG (difference-frequency-generation) laser absorption near 3.4 microns [56]. Representative *n*-heptane time-history results are shown in Fig. 13 along with simulations using different (±30%) *n*-heptane overall decomposition rate coefficients. Good sensitivity to the *n*-heptane decomposition rate is evident.

Rate coefficient data from both *n*-heptane experiments are shown in Fig. 14. Good agreement is observed between the two separate experiments and with the rate proposed by Davis and Law based on flow reactor and flame studies [57].

## 2.5. Oxidation systems

The importance to human society of understanding and controlling hydrocarbon oxidation kinetics cannot be overstated. Thus, one of the major goals of combustion research is to provide a reliable database of kinetic targets to enable the testing and validation of large reaction mechanisms and the development of simple correlations describing ignition and combustion of

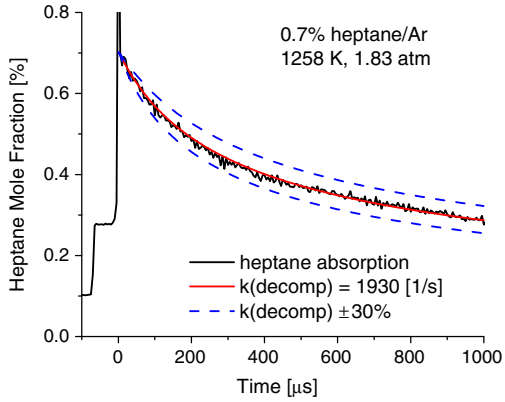


Fig. 13. *n*-Heptane time-history measurement during *n*-heptane decomposition.

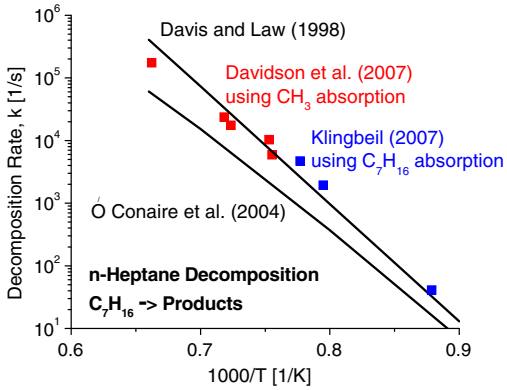


Fig. 14. Arrhenius diagram for the overall rate coefficient of *n*-heptane decomposition. Model rates from Refs. [57,39].

hydrocarbons. Shock tubes can provide three types of kinetic targets for this purpose: ignition delay times, species time-histories, and direct measurements of elementary reactions. We previously discussed the use of shock tubes to provide direct measurements of elementary reactions. Here we discuss the other two types of kinetic targets.

### 2.5.1. Example: ignition delay times, jet fuel surrogate components

Comparisons of ignition delay times with model simulations provide an overall indicator of mechanism performance. Correctly performed, ignition delay times are directly comparable from facility to facility. However, even though ignition delay times are relatively simple to measure in shock tubes, the wide scatter that has existed in these measurements from different laboratories in the past indicates that often, not all experimental variables have been properly accounted for. The key variables in a shock tube/constant volume measurement of ignition delay time are the temperature, pressure, mixture composition, and how closely the shock tube approaches constant–volume behavior. Temperature uncertainty, either from shock-speed measurement inaccuracy or from non-ideal behavior of the shock tube flows, is generally the most significant cause of scatter in shock tube ignition delay time measurements.

The development of simple surrogates for jet fuel requires kinetic targets for all of the surrogate components. Figure 15 presents the results of a series of high-pressure ignition delay time measurements for selected jet fuel surrogate components. Because of the high fuel concentrations (stoichiometric mixtures in air), these measurements were based on piezoelectric (PZT) pressure time-histories performed in a high-pressure 5 cm diameter shock tube at Stanford. At lower concentrations (i.e. hundreds or thousands of ppm of fuel) emission or laser absorption measurements can provide better results.

What is evident in Fig. 15 is the strikingly similar behavior at high temperatures (above 1000 K) of many of the alkane surrogate components. At lower temperatures, in the NTC (negative temperature coefficient) regime, larger differences are seen between the normal alkanes and other major components. Over all temperatures, the ignition delay times of toluene, an archetypal aromatic compound, are significantly longer than those of any of the alkane species.

### 2.5.2. Example: ignition delay times, high-pressure methane

Shock tube studies can also provide controlled measurements of ignition delay times and oxidation

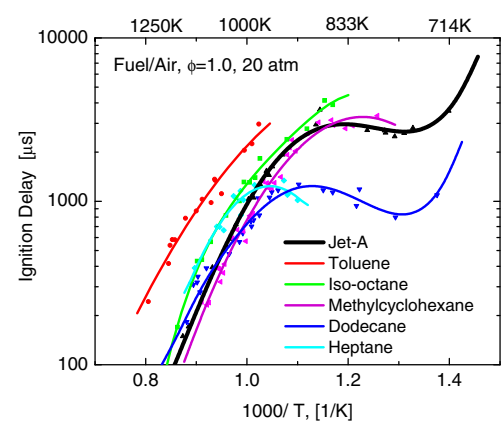


Fig. 15. Jet fuel surrogate ignition delay time measurements at 20 atm. Experimental data from Refs. [58–62]. Ignition delay time data presented here have not been corrected for gradual changes in the reflected shock pressures ($dP_5/dt$ effect). Equivalent constant-volume ignition delay time values for measurements longer than about 2 ms can be expected to be longer by 10–50% than those presented here.

rates at high pressures, representative of conditions found in practical devices, such as gun barrels and rocket engines. Ram accelerators, for example, rely on the ignition of high-pressure gaseous fuel behind projectile-generated shock waves to provide the needed thrust to reach ultrahigh velocities. Hence, ignition delay times of rich methane mixtures were critically needed to predict performance of these devices. To achieve these conditions, Petersen et al. [63,64] measured methane/oxygen/nitrogen ignition times at pressures from 40 to 260 atm using emission and pressure diagnostics in a high-pressure shock tube. Selected results are shown in Fig. 16. Species time-histories of OH (via UV laser absorption at 306 nm) and $CH_4$ (via emission near 3.4 μm) for similar conditions are given in Ref. [65].

*2.5.3. Example: multi-species time-histories, n-heptane oxidation*

Laser-absorption-based multi-species time-histories have the potential to provide kinetic information for several different classes of species: reactants, small-radical species, stable intermediate species, and combustion products. Figure 17 shows a recent application of this strategy to the study of *n*-heptane oxidation at 1500 K by Davidson et al. [66] Here, *n*-heptane and ethylene were monitored using IR gas laser absorption at 3.39 and 10.53 μm, respectively; OH was monitored using UV laser absorption at 306.5 nm; and $CO_2$ and $H_2O$ were monitored using tunable IR diode laser absorption at 2.7 and 2.5 microns, respectively. Data are also available at a slightly different fuel concentration for $CH_3$ [55].

In Figure 17, several time scales and reaction intervals for the different species are evident. At early times up to about 10 μs, fuel decomposition rates and pathways determine the stable-intermediate and small-radical concentrations. Here, *n*-heptane time-history measurements, combined with measurement of the major decomposition product ethylene, provide strong constraints on the reaction rates and pathways used in the reaction mechanism; many of these rates and pathways are often only estimated and have not been measured [67]. During the ignition induction delay from approximately 10–100 μs, plateau levels of the species OH provide information on the small-radical pool (H- and O-atoms). These atomic species are difficult to measure directly, but their concentrations are intimately related to the OH concentration time-history by the H/O reaction set that rapidly equilibrates these small-radical populations. Current mechanisms do not yet capture the correct OH time-history behavior at early times, and this is likely related to uncertainties in the rate coefficients of the reactions of OH with intermediate *n*-heptane composition products. The overall rate of oxidation is indicated by formation of $H_2O$ (measured here) and CO (not measured here). Measurement of both a reaction progress marker, such as $H_2O$, and a representative small-radical species responsible for radical branching, such as OH, places strong constraints on the rates of oxidation processes that are occurring during the ignition induction delay. Constraints on the post-ignition behavior of the system can be derived from the time-history of $CO_2$, which continues to form at longer times.

Experiments of this type, in which multiple species are monitored by laser absorption, offer exciting new potential for refining and validating detailed mechanisms. In support of this strategy, researchers are currently working to expand the set of species that can be monitored. Advances in infrared diode laser absorption methods, in particular, continue to provide new access to wavelengths that can discriminate, as much as possible, individual $C_4$ and higher hydrocarbons.

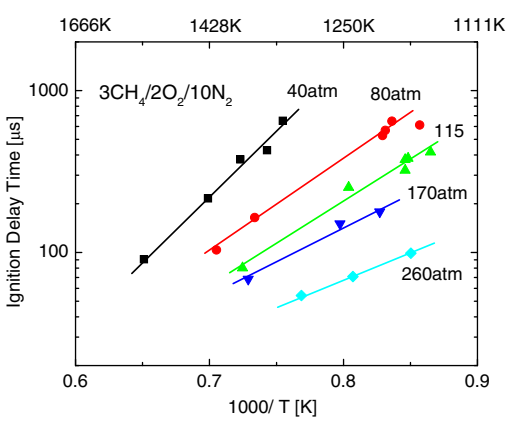


Fig. 16. High-pressure methane/oxygen/nitrogen ignition delay time measurements in reflected-shock experiments. Experimental data from Ref. [63].

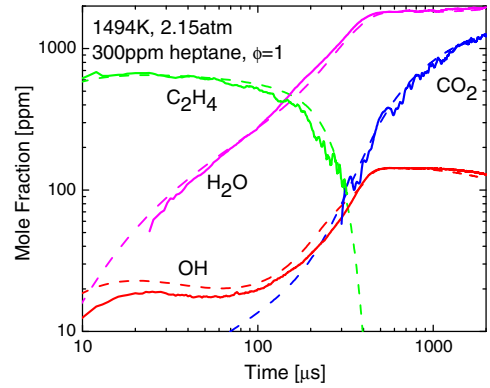


Fig. 17. Multi-species time-history measurements of *n*-heptane oxidation [66]. Dashed lines are Sirjean et al. (2009)/JetSurF 1.0 simulations [68].

### 2.6. Pollutant kinetics

Accurate rates for the Zeldovich reactions [69] are essential to accurate simulations of high-temperature formation of $NO_x$ (nitrogen oxides) pollutants. Interestingly, perhaps the first application of laser absorption diagnostics to combustion kinetics in shock tubes was performed on this key reaction system, 35 years ago, by Monat et al., who measured the rate coefficient for $N_2 + O \rightarrow NO + N$ using absorption of CO laser light at 5.17 μm to monitor formation of NO [70]. Measurements of other $NO_x$ reactions, e.g. $O + NO \rightarrow N + O_2$ and $NO + NH_2 \rightarrow$ products, using this pioneering diagnostic technique, followed [71]. These early measurements were later extended to monitor $NH_2$ at 597 nm using a CW ring dye laser and frequency modulation (FM) absorption spectroscopy to achieve accurate measurements of the important branching ratio of the products of $NO + NH_2$ [72].

Recently there has been renewed interest in fuel-nitrogen chemistry, particularly in establishing the products of the prompt NO reaction of $CH + N_2 \rightarrow$ products. Lin and co-workers have suggested that the primary initial products are (a) $H + HCN$, rather than (b) $HCN + N$ [73]. Validation of this hypothesis required measurement of the overall reaction rate as well as confirmation of the product NCN. Vasudevan et al. were able to do this using laser absorption measurements of both CH and NCN in shock tube experiments [74].

The overall reaction rate coefficient and the branching ratio were measured using CW, narrow-linewidth laser absorption of CH at 431.1 nm. First, the overall rate coefficient was inferred over the temperature range of 1943–3542 K by kinetically modeling CH profiles in shock-heated dilute mixtures of ethane (or acetic anhydride) in argon with variable addition of $N_2$. Figure 18 shows example data from this experiment and the strong sensitivity of the CH profile to the addition of $N_2$. Subsequently, the branching ratio was inferred over the temperature range of 2228–2905 K from experiments in dilute mixtures of ethane in pure nitrogen. These measurements were found to be consistent with a branching ratio $k_a/(k_a + k_b)$ near 1, establishing $NCN + H$ as the primary products of the $CH + N_2$ reaction.

In separate experiments, providing another example of the power of species-specific laser absorption, the production of NCN was confirmed by CW narrow-linewidth absorption. Spectroscopic confirmation of the formation of NCN can be seen in Fig. 19. Here the peak absorption levels occurring during reflected shock waves in ethane/$N_2$ mixtures were found to vary as a function of wavelength near 329 nm, consistent with occurrence of the NCN $\Pi-\Sigma$ bandhead reported by Smith based on observations in flames [76]. The optimum (peak) absorption wavelength for NCN detection was thus established as 329.13 nm. Figure 20 shows comparison of a measured NCN absorption record with a modeled NCN time-history. The experimental absorption can be converted to mole fraction using a best-fit (at this temperature) absorption coefficient of $k_{329} = 58\ \text{atm}^{-1}\ \text{cm}^{-1}$. This comparison technique provides a method to determine the absorption coefficient for transient species (such as NCN) where more absolute methods are not available. Note also the good agreement between the shape of the measured and simulated NCN time-history providing additional confirmation of the detailed kinetic model with a branching ratio of 1.

The experiments reported above, and others, provide tangible evidence of the important and growing role of species-specific laser absorption methods in modern shock tube studies of combustion kinetics.

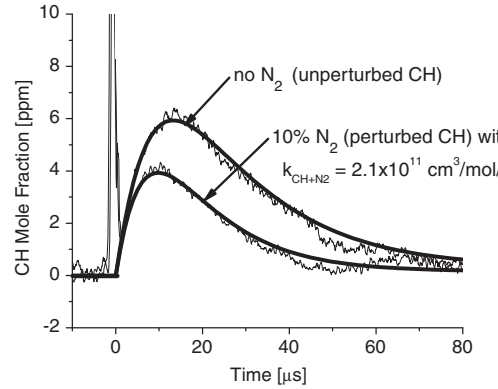


Fig. 18. Laser absorption by CH at 431.1 nm. Initial conditions: 3348 K, 0.95 atm, 10 ppm ethane in argon with and without the addition of 10% $N_2$. Data from Ref. [75].

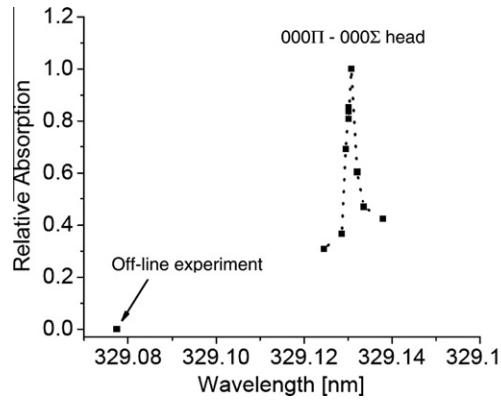


Fig. 19. Relative peak absorption measurements at 2775 K, 0.6 atm, 113 ppm ethane/$N_2$. Laser data from Ref. [75].

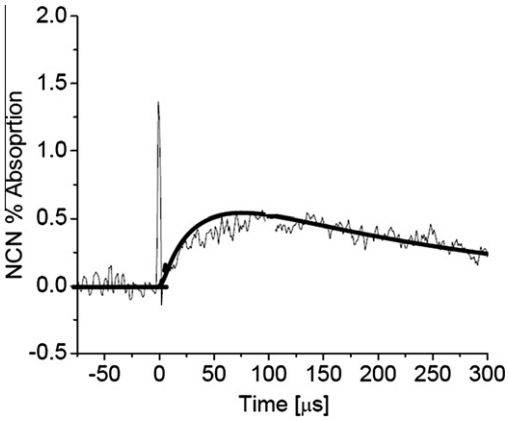


Fig. 20. Determination of NCN absorption coefficient $k_{329}$ by comparison of measured and simulated NCN time-histories. Initial reflected-shock conditions: 2495 K (equilibrated), 0.45 atm, 105 ppm ethane, 9.8% He, balance $N_2$. Comparison of experiment shown as thin solid line, with simulation shown as heavy solid line (see Ref. [74] for mechanism details), finds that for these conditions, the best-fit value of $k_{329} = 58\ \text{atm}^{-1}\ \text{cm}^{-1}$.

### 2.7. Advances in shock tube operation

There is a continuing need to improve shock tube performance. To reduce uncertainties in measurements of reactive systems with large activation energies or strong pressure sensitivities, more uniform temperature and pressure test conditions are required as well as improved means of accurately monitoring these temperatures and pressures. Such shock tube improvements can be achieved by reducing (and in some cases effectively eliminating) non-ideal facility effects through the addition of driver-section inserts and the use of driven-section buffer gas volumes [77].

To extend shock tube measurements to include the NTC (negative temperature coefficient) ignition regime or cool flame conditions, longer and more uniform test times are required. This can be accomplished with driver-section extensions and tailored gas mixtures. Recently, extended shock tube test times to greater than 35 ms have been demonstrated [78].

To study JP-series fuels, diesel fuels, kerosene fuels, and their surrogates, and heavy hydrocarbon components, such as *n*-dodecane or *n*-hexadecane, without significant facility heating (i.e. greater than about 150 °C or higher), a new method for handling low-vapor-pressure fuels is required. This has been accomplished recently by development of an aerosol shock tube that enables the study of low-vapor-pressure fuels without fuel cracking by pre-heating [79].

Here we discuss two of these advancements: the use of driver inserts to produce uniform test conditions (and laser-based temperature sensing to validate this performance) and the development of the aerosol shock tube, both based on work at Stanford.

#### 2.7.1. Example: uniformity of test conditions: the use of driver inserts

Non-ideal shock tube facility effects, such as diaphragm bursting dynamics, incident shock wave attenuation, and boundary layer growth behind the incident and reflected shock waves, can cause variations in the pressure time-histories seen in reflected shock wave experiments. These variations can be minimized or eliminated by the use of driver inserts [77]. Driver inserts, or other forms of change in the driver section diameter, are sources for expansion waves that are able to counteract the gradual increases generally seen in reflected shock pressure profiles. As the expansion wave progresses into the driver section from the burst diaphragm station, it is partially reflected from the driver insert and travels the full length of the shock tube driven section, weakening the post-reflected-shock pressure profile [80].

With these driver inserts, near-constant-volume performance can be achieved behind the reflected shock wave. This near-ideal behavior simplifies the interpretation of shock tube kinetics experiments, particularly in experiments that are highly sensitive to temperature and pressure changes, such as measurements of ignition delay time of hydrocarbon fuels.

Figure 21 shows the effect of the insert on a shock wave in pure argon. The pressure profiles shown, measured 2 cm upstream from the shock tube end wall, show the jump in pressure to the reflected shock condition (4 atm). Without the driver insert, the reflected shock pressure gradually increases to about 4.5 atm. With the driver insert in place, the pressure profile remains nearly constant, effectively maintaining a constant-volume test gas condition.

Accurate temperature measurements in shock tube flows are difficult to perform, in part because of the short time frames involved and the need for absolute calibration at high temperatures. However, recently Farooq et al. have developed a method to measure temperature using a sensitive form of absorption known as wavelength modulation spectroscopy, with $CO_2$ as the species sensed. [81]. Using this method, the temperature time-history of the test gas volume behind the reflected shock wave was measured. Exceedingly uniform temperatures were shown to accompany the uniform pressures present with the use of the driver inserts. Representative measurements are shown in Fig. 22. We expect this new temperature diagnostic will find extensive use in future shock tube kinetics studies where temperature varies due to chemical reac-

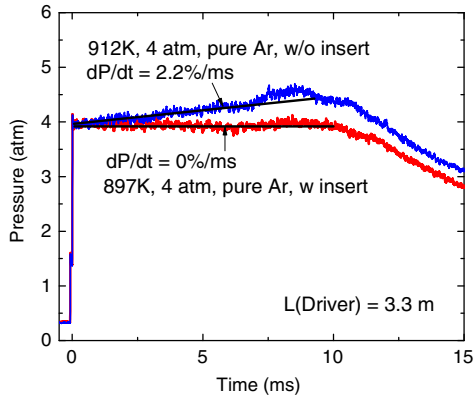


Fig. 21. Effect of the driver insert on pressure profile in Stanford shock $dP/dt$ (expressed in %/ms) behind the reflected shock wave is reduced from 2.2%/ms to 0%/ms with the addition of the driver insert.

tions and accurate measurements of temperature change may provide new means of evaluating kinetic mechanisms and rates.

*2.7.2. Example: aerosol shock tube and distillate fuels*

Kinetics studies in shock tubes have primarily been limited to gas-phase mixtures. This is partially due to the difficulty of uniformly distributing an aerosol throughout the shock tube test volume. Recently, a method was developed at Stanford to introduce uniform aerosol mixtures of small droplets into a shock tube. Using this new aerosol shock tube it has been possible to study low-vapor-pressure fuels such as diesel and jet fuel that are difficult to produce in the purely vapor phase. With this approach, it is not necessary to pre-heat the fuel for evaporation. In the aerosol shock tube that is done behind the incident shock wave, and in a way that is representative of droplet evaporation in the fuel spray of a practical combustor. Thus, there is little concern about fuel decomposition or oxidation because of pre-heating, nor is there concern about fractional distillation of distillate fuels that would occur when pre-experimental heating is used to vaporize the fuel [79]. A schematic of the current (second-generation) aerosol shock tube is shown in Fig. 23.

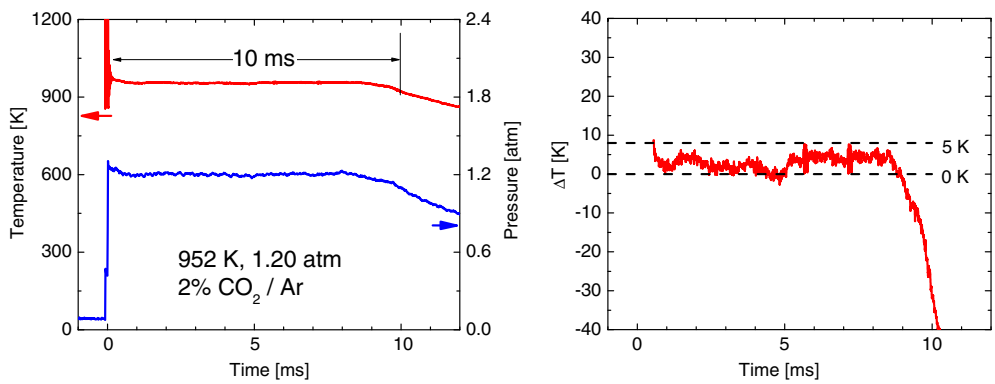


Fig. 22. Accurate determination of reflected shock temperature using $CO_2$ WMS (wavelength modulation spectroscopy) at 2.7 μm. A highly-uniform temperature profile is obtained when pressure is maintained constant using a driver insert. Driver gas helium/nitrogen (60/40%), driven gas argon seeded with 2% $CO_2$. Reflected shock conditions: $T_5 = 952$ K, $P_5 = 1.2$ atm. Figure adapted from Ref. [81].

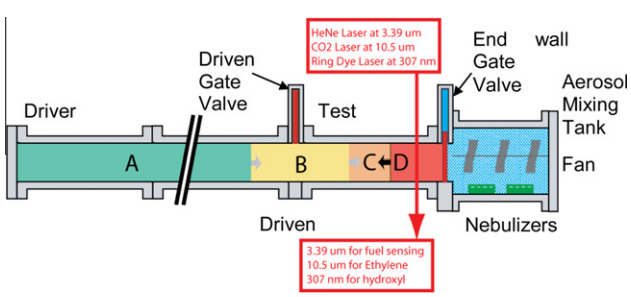


Fig. 23. Schematic of aerosol shock tube (AST II) setup at Stanford. The different regions in the tube are (from left to right): (A) driver gas, (B) incident-shock-heated driven gas with no fuel, (C) incident-shock-heated test mixture with fuel, and (D) reflected-shock-heated test mixture.

The operation of the aerosol shock tube is conceptually straightforward. A well-stirred mixture of fuel aerosol and carrier gas is generated in the aerosol mixing tank with ultrasonic nebulizers. This aerosol is then drawn into a constrained test section at the end of the shock tube as plug flow. The constraining valve (i.e. the driven gate valve) is then opened and the diaphragm quickly burst allowing an incident shock wave to travel through an effectively uniform aerosol test mixture. The higher temperature and pressure conditions behind the incident shock wave rapidly vaporize the fuel, and the reflected shock wave then raises this vaporized test gas mixture to the final desired conditions.

With this aerosol shock tube, it is possible to study fuels with very low vapor pressures including commercial diesel (DF-2) and heavy surrogate components such as *n*-dodecane and *n*-hexadecane [82]. Representative measurements of ignition delay times for diesel fuel, and *n*-hexadecane, a diesel surrogate component, are shown in Fig. 24. Measurements at these high fuel concentrations, and for these fuels, have not been feasible in conventional shock tubes.

### 2.8. Advances in shock tube modeling

#### 2.8.1. Example: CHEMSHOCK

Validation of detailed reaction mechanisms to describe combustion processes relies on direct comparison of mechanism simulations with reliable experimental measurements. However, even for two of the simplest hydrocarbon-related combustion systems, hydrogen and propane oxidation, large uncertainties have existed in the abilities of current detailed mechanisms and gas dynamic models to accurately simulate ignition delay time and species time-histories in shock tube experiments conducted at lower temperatures.

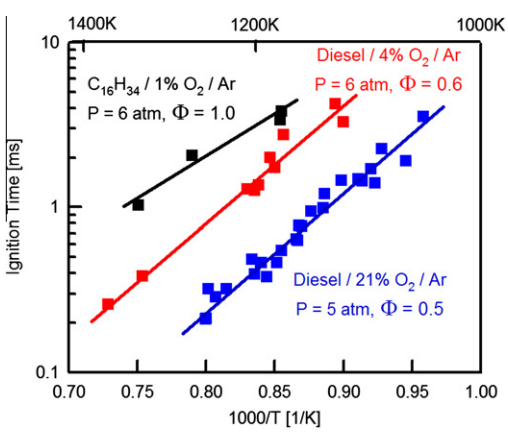


Fig. 24. Ignition delay times for diesel and *n*-hexadecane [83]. All data normalized to common conditions using $P^{-1}$ and $\varphi^{1}$ scaling.

These uncertainties have been linked to uncertainties in the detailed reaction mechanism at low temperatures (below 1000 K) where hydroperoxyl reactions are believed to be important, and to uncertainties introduced by non-ideal facility-related effects. These facility effects typically become important only at long test times (greater than few ms). At shorter test times, large diameter shock tubes are relatively immune to non-ideal problems, especially when relatively dilute fuel-oxidizer mixtures are employed.

To understand the effect of non-constant experimental conditions, a new model for reactive gas dynamics behind reflected shock waves has been developed and a zero-dimensional code assembled based on this model: CHEMSHOCK [84]. In contrast with past reflected-shock modeling based on assumptions of constant volume and internal energy, CHEMSHOCK accounts directly for the effects of changing pressure and temperature (facility-related and chemistry-related) through use of the experimental pressure history.

The CHEMSHOCK model assumes a two-step process over each small time step: (1) reaction by the normal constant U, V process; followed by (2) an assumed isentropic adjustment in conditions to recover the measured value of pressure at this time. The CHEMSHOCK model more accurately models the reflected-shock ignition process than the traditional approach using CHEMKIN with a constant U, V constraint, in that the new approach allows for facility-dependent effects and energy-release phenomena in the reflected-shock environment. This approach is valid up to the time of ignition when the pressure rises by a large amount and the flow may transition to a detonation or become three-dimensional [85]. In its present form, CHEMSHOCK has been applied only to the common case in which the earliest ignition occurs adjacent to the end wall, but work is in progress to extend the model to the more general one-dimensional case of non-uniform reaction in the reflected region.

The utility of this new gasdynamic model, as well as the advantages of the driver insert model have recently been demonstrated in low-temperature, high-pressure propane ignition delay time measurements by Lam et al. [86]. These data are shown in Fig. 25. Measurements of ignition delay times for propane were made with and without driver inserts. This allows comparisons of ignition delay times for two cases: those with a near-constant-volume reactor constraint with no induced pressure gradient, i.e. $dP_5/dt = 0\%/\text{ms}$, and those with a large facility-induced pressure gradient $dP_5/dt = 6\text{–}7\%/\text{ms}$. The near-constant-volume ignition delay times provide a database for low-temperature propane model development that is independent of non-ideal fluid flow and heat transfer effects. Comparison of these

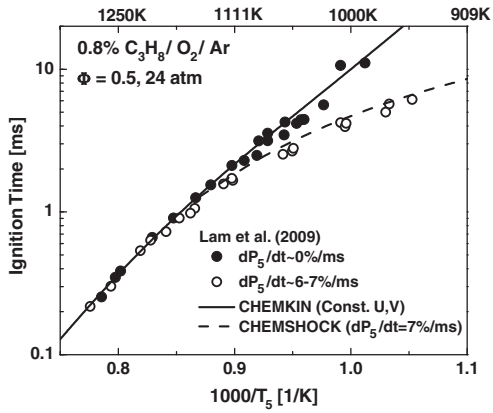


Fig. 25. 24 atm propane ignition data [86] and CHEMKIN ($dP/dt = 0$, constant volume constraint) and CHEMSHOCK (using experimental pressure variation, $dP/dt = 6$–$7\%$/ms) modeling. Both simulations use the Sirjean et al. (2009)/JetSurF v1.0 mechanism [68].

near-constant-volume measurements with predictions using the Sirjean et al. (2009)/JetSurF v1.0 mechanism [68] show good agreement at all temperatures [77]. Ignition delay times measured with $dP_5/dt = 6$–$7\%$/ms were found to be significantly shorter (about 1/3 of the near-constant-volume values) at the lowest temperatures. However, these ignition times are successfully simulated using the same detailed mechanism when CHEMSHOCK is used to account for changes in pressure and temperature. These results serve to confirm the JetSurF kinetic model while also illustrating the importance of proper modeling of facility gasdynamic effects.

### 2.9. Future of laser absorption/shock tube kinetics studies

#### 2.9.1. Accomplishments

Shock tubes have historically been a primary source of combustion kinetics data at elevated temperatures. The implementation of laser absorption diagnostics with high sensitivity and accuracy has enabled improved shock tube experiments, leading to more accurate rate coefficient data for elementary reactions as well as unique target data sets needed to evaluate and refine detailed kinetic mechanisms for pyrolysis, ignition and oxidation of relevant fuels and fuel surrogates. Recent improvements in shock tube facilities (use of tailored driver gas mixtures, extension of driver lengths, and driver inserts) have extended experimental test times, enabling experimentation at lower temperatures where ignition delay times and rate coefficient data are lacking, while also reducing uncertainty and error in reaction conditions, thereby providing more accurate determination of reaction rates. Advances in modeling of reacting shock tube flows have also contributed to improvements in the interpretation of shock tube kinetics data.

#### 2.9.2. Needs and opportunities

Many aspects of shock tube/laser absorption kinetics provide future research opportunities. This includes improvements in both shock tubes and laser diagnostics, as well as targeted applications in hydrocarbon kinetics.

There continues to be a need for improved shock tube modeling that can properly account for energy changes behind reflected shock waves. New multi-dimensional gasdynamic models should offer more accurate simulations of shock tube/laser absorption experiments than are currently achieved with constant U, V or constant H, P models, as well as extending applicability to times beyond ignition.

There continues to be a need for development of laser absorption diagnostics for new species, particularly for use in multi-species time-history strategies. One measure of success in this area will be the ability to begin to close the carbon balance through measurements of multiple hydrocarbon intermediates. This will require new IR diagnostic capability to measure higher alkenes such as propene, 1- and 2-butene, and 1,3-butadiene, and simple alkanes such as methane and ethane. Similarly, the extension of current UV capabilities for transient radical species to larger species such as propargyl, ethyl and propyl radicals, will enable target data with stronger constraints on modeling decomposition pathways for large fuel components.

Many critical elementary reactions important in modeling hydrocarbon oxidation have not been accurately measured. For example, rate measurements are needed for several key reactions involved with peroxy ($HO_2$) radicals, including $CH_3 + HO_2 \rightarrow$ products, $H + O_2 + M \rightarrow HO_2 + M$ for the major syn–gas components M=CO, $H_2O$ and $CO_2$, and alkyl $+ O_2 \rightarrow$ alkene $+ HO_2$. Measurements of these $HO_2$ radical reactions by spectroscopic detection are challenging owing to interferences in the UV absorption spectra by some of the accompanying reactant (and product) species.

Measurements are needed for decomposition products and pathways for larger diesel fuel surrogates including hexadecane and bio-diesel compounds. The aerosol shock tube strategy, coupled with laser absorption diagnostics, should provide access to these heavier fuel components. Studies of cyclo-alkanes, including methylcyclohexane, cyclohexane, and butylcyclohexane, as well as of polycyclic species, both paraffinic and aromatic, are also needed.

And finally, there is a need and an opportunity to develop multi-species time-history databases. Initial work in this area on *n*-alkanes is now in

progress [19,66,83], however data are still needed for the NTC (negative temperature coefficient) regime for *n*-alkanes such as *n*-dodecane to better understand the first-stage ignition process. Equally important are data sets for aromatics including toluene and alkyl-benzenes, such as propylbenzene, that are major components of jet fuel. We anticipate that high-accuracy species time-history databases from shock tube experiments will become an important standard for mechanism and model validation in the future.

## 3. Propulsion

The development of new propulsion concepts has, in recent years, come to rely on modern laser diagnostic tools to characterize ground test facilities and to test and validate computational models for the complex chemically reacting flowfields associated with high-speed propulsion. Here, following a brief review of diagnostics fundamentals, five example applications of laser diagnostics to propulsion problems are reviewed. This discussion will not include laser diagnostics in internal combustion engines, which have been reviewed in two recent combustion symposium plenary papers [87,88]. Similarly discussion is limited to diagnostics that are species-selective by tuning the laser wavelength to match absorption transitions in the target species, i.e. absorption and fluorescence (laser-induced fluorescence or LIF). Thus, important laser-light-scattering combustion diagnostics such Rayleigh, Raman, and particle imaging velocimetry (PIV) will not be included; nor will we discuss the coherent non-linear variants of these scattering diagnostics (e.g., coherent anti-Stokes Raman scattering), and will instead direct the interested reader to comprehensive reviews and the references therein [1,89,90].

In the first example, we discuss the use of spatially resolved planar laser-induced fluorescence (PLIF) to investigate the high-speed mixing needed to stabilize air-breathing combustion in supersonic combustion ramjets (i.e. scramjets). Then, in four subsequent examples, the application of laser-absorption diagnostics discussed previously for chemical kinetics studies in shock tubes is extended to the use of tunable diode lasers and fiber optics for sensing in scramjet combustors, pulse detonation engines, gas turbine combustors, and hypersonic test facility characterization. We illustrate how line-of-sight absorption measurements using reliable solid-state diode lasers can overcome their lack of spatial resolution, and through appropriate sensor design can provide the information needed to optimize and control the performance of new propulsion concepts. The low power consumption of these diode lasers and the robust optical engineering enabled with fiber optics give these diagnostics the potential to transition to sensors for in-flight closed-loop control.

### 3.1. Diagnostics fundamentals

#### 3.1.1. Absorption

Absorption diagnostics are line-of-sight and thus ideally suited for applications with near-uniform temperature, pressure and gas composition, e.g., the flows associated with many high-speed-propulsion test facilities. Thus, some of the first applications of absorption diagnostics to propulsion research have been to characterize the flow in ground test facilities [91–93]. The development of wavelength-tunable diode lasers and associated fiber optics technology in the near-infrared by the telecommunications industry has provided low-cost, low-power-consumption, laser sources with excellent potential for gas sensing. Both primary combustion products $CO_2$ and $H_2O$ have rich absorption spectra in the near- and extended-near-infrared as illustrated in Fig. 26 for a typical combustion temperature of 1500 K. Although the overtone and combination bands appear dense due to the large number of transitions, individual transitions can be spectrally resolved to facilitate quantitative measurements using Beer's law as formulated in Eq. (1). The strength of the absorption lines depends on the thermal population of the specific rotational–vibrational absorbing state (a function of the amount of internal energy $E''$ of that level). Therefore the ratio of two absorption signals from lines of the same species with different $E''$ depends only on gas temperature, and indeed the use of two-line water vapor absorption for temperature measurements has become routine for combustion and propulsion flows. The absorption lineshapes are primarily broadened by collisions (pressure), providing an opportunity for pressure measurements by measuring the absorption lineshape and accounting for the temperature-dependent Doppler contribution to the lineshape. Fortunately, for many of these near-infrared $H_2O$ and $CO_2$ transitions the broadening parameters are tabulated in HITRAN for

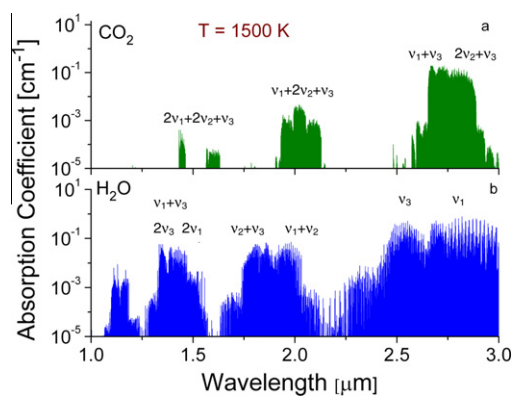


Fig. 26. Absorption coefficients from 1–3 um for $CO_2$ (a) and $H_2O$ (b) at 1500 K from HITRAN [36].

self-broadening and broadening by air [36], though they typically must be measured as part of any sensor development effort involving high-temperature applications and for other broadening species, as very little high-temperature broadening data exists in the literature. At present, the most accurate values for line positions and strengths can be found in the recently released 2010 update of HITEMP [36]. The center wavelength of an absorption lineshape in a moving flow is Doppler-shifted, and the differential line-center shift between laser light directed upstream and downstream in the flow provides a convenient sensor for velocity [94–100]. In the applications of absorption sensors to propulsion ground test discussed below, all of these phenomena will be used to monitor the chemically reacting flows.

Absorption measurements are called direct absorption when the narrow-linewidth laser is either fixed at the line-center of the feature or is scanned in wavelength across the feature. Scanned-wavelength direct absorption has the advantage that the absorption signal can be integrated over the feature, thereby eliminating dependence on the pressure- and Doppler-broadened lineshape while providing a quantitative, calibration-free, measurement. By contrast, fixed-wavelength absorption on line-center must account for any changes in the lineshape function. For small absorption levels, the accuracy of direct absorption measurements is limited by the determination of the "no-absorption" baseline of laser intensity ($I_0$ in Eq. (1)). Typically this baseline is determined by scanning the laser far enough on either side of the transition to "fit" the baseline at line-center, or for fixed-wavelength absorption by evacuating the measurement volume, which severely limits the fixed-wavelength strategy in ground-test facilities with short test times. The measurement bandwidth of the fixed-wavelength approach can be extremely large and is limited only by the time-response of the detector and data recorder (half the ADC sampling rate). Although the bandwidth of scanned-wavelength absorption is limited by the rate of tuning the laser far enough beyond the absorption transition to recover the off-absorption baseline, modern diode lasers have allowed measurements of $H_2O$ and $CO_2$ at several 10s of kHz.

For small absorption, the signal given by Beer's law becomes the small difference between two relatively large measured intensities. It has long been known that the sensitivity of absorption measurements can be improved by rapidly modulating the laser wavelength and measuring the absorption signal at harmonics of the modulation frequency f, and for many practical diode laser absorption applications, wavelength modulation with second-harmonic detection (WMS-2f) is the method of choice. [95,101–104]. Injection-current-tuned diode lasers have synchronous amplitude and laser-frequency modulation. Thus, the 1f harmonic of the transmitted laser intensity can be used to normalize the WMS-2f signal and account for the laser-intensity dependence of the WMS-2f signal; this fact has long been recognized but has been independently discovered by numerous researchers [105–110]. For practical measurements in propulsion and in large-scale combustion systems, this normalization conveniently accounts for window fouling and losses in the laser intensity from non-absorption (e.g., scattering) sources. WMS-2f signals are amenable to real-time data interpretation and this fast processing of WMS-2f signals facilitates closed-loop control using diode laser sensors. When the laser modulation characteristics are characterized in the laboratory before employing a sensor, this normalization can provide calibration-free WMS-2f/1f [108–110] rendering absorption-based sensing practical in a wide variety of harsh environments.

Absorption measurements on a common line-of-sight by multiple diode lasers (i.e., wavelength multiplexing) have been facilitated in the near-infrared by the large telecommunication investment, and such architecture has been exploited in laser sensors for more than a decade [111]. The use of fibers eliminates operator adjustments to carefully overlap laser beams and facilitates delivery of laser light over distance to remote facilities and hostile environments. Three different de-multiplexing strategies have been used for the transmitted laser intensity. Time-multiplexing simply powers the lasers one-at-a-time, which can be done at up to MHz rates. Only one detector is needed, but for scanned-wavelength applications, the switching rate can be slow enough that the advantage of a common beam path can be lost. WMS sensors can also use a single detector and demultiplex the electronic signal by modulating the lasers at different frequencies. This works quite well for two lasers, but separating the signal harmonics for more than two lasers can also restrict the measurement bandwidth. Lastly, the co-propagating transmitted laser beams can be separated onto multiple detectors using spectral dispersion, e.g. with a grating or even a prism. This has the advantage of retaining the maximum measurement bandwidth, but requires well-separated laser wavelengths. Hybrid approaches, combining frequency- and spectral-demultiplexing are also employed [112].

### 3.1.2. Planar laser-induced fluorescence

Absorption of a laser photon tuned to a specific wavelength (transition) is the first step in laser-induced fluorescence (LIF) (see Fig. 27). The absorbing molecule or atom is typically excited with visible or ultraviolet light into an excited electronic state coupled to the ground state with one-photon selection rules (although LIF variants exist with excited electronic states

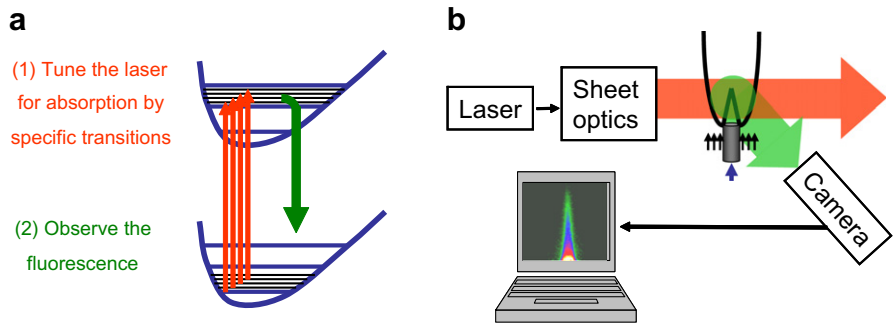


Fig. 27. (a) Diatomic molecule energy levels for LIF with the upward arrows showing the absorption of rovibrational-specific electronic transitions and the downward arrow depicting the fluorescence signal. (b) PLIF with laser beam made into a sheet of light, directed through the flame and collected onto an ICCD camera; signal on laptop from IR-PLIF of CO in the fuel-rich core of a Bunsen flame [114].

coupled with multiple (usually two) photons). The cartoon in the left panel of Fig. 27 illustrates excitation of a diatomic molecule by one-photon absorption. The electronically excited molecules can subsequently decay by emission of a photon (fluorescence) or non-radiatively via collisions or intramolecular transfer to other electronic states. The fluorescence emission is the detected LIF signal. The fluorescence lifetime of the excited state is usually dominated by collisional quenching and is typically a few nanoseconds at atmospheric pressure. Thus, if the excitation uses pulsed lasers with a pulse length of a few ns or less, even high-speed flow is effectively frozen, and spatially resolved LIF can provide an instantaneous snapshot of the flowfield structure [89,90,113].

Expanding the laser beam into a flat sheet of light enables the fluorescence signal to be imaged onto a camera, as illustrated in the right panel of Fig. 27. This approach is called planar laser-induced fluorescence (PLIF) imaging and provides a two-dimensional visualization of target species concentration distribution in the flowfield. Assuming the use of a single short laser pulse with rapid fluorescence, this diagnostic method is highly quantitative with regard to space and time, providing accurate, time-resolved flowfield-structure information. Interpretation of the signal in terms of species concentration is often semi-quantitative though PLIF thermometry based on ratios of two images can be quantitative [115,116]. Measurements of NO and temperature in a model scramjet flow in a shock tube provided the first "instantaneous" two-color PLIF imaging [117].

The first PLIF imaging was reported for OH in simple flames nearly thirty years ago [118,119], and since then PLIF has become a ubiquitous diagnostic for combustion research. PLIF imaging can also be performed using vibrational excitation in the infrared, rather than electronic excitation, as demonstrated in the cartoon display in the right panel of Fig. 27 showing a PLIF image of CO in a Bunsen flame [114]. The research to develop and apply PLIF to combustion problems is so vast and varied that it is impossible to adequately review all of the discoveries or cite all the relevant research here, and instead we will direct the reader to three exemplary reviews and the many references cited therein [89,90,113].

During the past decade much combustion diagnostics research has focused on the transition of PLIF diagnostics to high pressures [120,121]. For example, quantitative measurements of NO and $CO_2$ have been reported up to 60 bar in premixed laboratory flames [122–124] and measure-

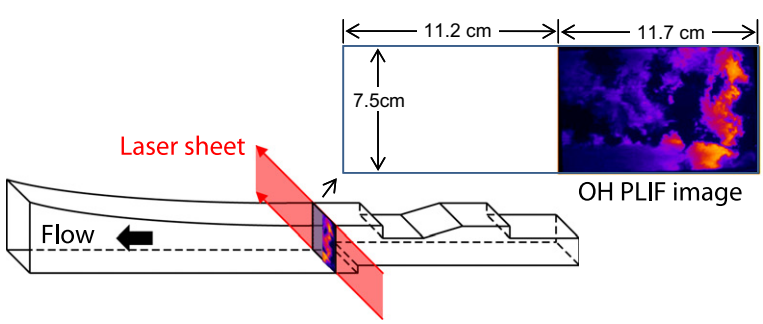


Fig. 28. Single-shot OH PLIF image from scramjet combustor ground test (see Refs. [142,143]).

ments of OH at 63 bar reported for trans-critical hydrogen–liquid oxygen flames [125]. These high-pressure PLIF techniques are beginning to transition from diagnostics research to diagnostic tools, and several applications of PLIF have been reported in model or modified gas turbines (e.g., Dreizler and Janicka [126] cite 30 examples in 2002). The complexity of gas turbine applications includes the presence of fuel spray [127] as well as some of the highest pressures, e.g. OH PLIF was performed at 20 bar in a sector-rig combustor [128]. In addition, PLIF of OH in high-pressure cryogenic liquid-oxygen/hydrogen jet flames has been used to validate model calculations on flame stability [129].

Another current research frontier of PLIF diagnostics is the development of high-speed imaging. PLIF imaging typically has been limited to imaging rates from 1 to 20 Hz by the repetition rates of commercially available pulsed lasers. The earliest high-speed imaging synchronized multiple lasers with multiple cameras, e.g., see reference [130]. Recently, good progress has been made in high-speed camera systems (intensified and unintensified), now available from multiple vendors, capable of imaging (at variable pixel resolution) from 1 to 100 kHz. During the past few years, significant advances in high-speed fluorescence imaging have been reported by at least four different groups, using either excitation lasers based on new solid-state technology or a pulse burst scheme: (1) PLIF imaging of OH at 5 kHz using a custom-built tunable dye laser pumped by a high-repetition-rate Nd:YLF laser, and a commercial (but specialized) high-speed intensified CMOS camera [131,132]; (2) PLIF imaging of biacetyl at 12 kHz using a frequency-tripled YAG laser at 355 nm, also with a commercial intensified camera [133]; (3) combined PLIF of OH and PIV [134]; (4) ultra-fast imaging (100 kHz) of NO using a novel burst-mode YAG laser (limited to pulse bursts at 1–10 Hz rates) to pump an optical parametric oscillator and a special camera [135–137]. Recently, we have begun to develop an alternative high-speed PLIF imaging strategy that uses continuous excitation at 266 nm with cavity doubling of a commercial high-power frequency-doubled CW-YAG laser. Short-time-gate PLIF of toluene was demonstrated in an expanding jet showing feasibility of this alternative for high-speed PLIF [138].

### 3.2. Ignition / flame holding in scramjet ground testing

Discussion is focused here not on diagnostics research but on the use of PLIF to study combustion phenomena; the example selected is flame holding for hypersonic air-breathing scramjets. In this application, the residence time in a combustor must be sufficient to allow for mixing of the fuel and air followed by complete combustion. This presents a significant design challenge, and the common solutions anchor the flame in lower velocity regions, in wall cavities or behind flow obstructions or steps [139]. Pioneering applications of PLIF imaging in model scramjet flows were performed by McMillin et al. [140] using NO as a tracer and by Ben-Yakar et al. [141] imaging OH. The evaluation of hypersonic combustor/fuel-injector designs remains an active area of research and PLIF is an ideal tool to investigate the flame structures in ground-test facilities. The OH radical is an important flame intermediate species that is ubiquitous in air-breathing hydrocarbon propulsion marking high-temperature regions where the flame is and where it has been. Thus, OH PLIF has become a commonly employed tool to evaluate fuel injector design schemes to enhance flow mixing. Here we highlight a series of PLIF experiments performed on hydrocarbon-fueled scramjets in a direct-connect hypersonic tunnel at the US Air Force Research Laboratory at Wright Patterson, OH [142,143]. PLIF of OH was performed to empirically improve combustor hardware design as well as to develop semi-quantitative understanding of the flame holding. Figure 28 shows a schematic of the flow path showing the location of the sheet of excitation laser light just downstream of the steps in the walls. The OH radical was excited with frequency-doubled dye laser light near 283.5 nm, which excites the $Q_1(8)$ transition of the $A^2\Sigma^+ \leftarrow X^2\Pi\ (v' = 1,\ v'' = 0)$ band of OH, and the light was collected onto an intensified CCD camera. A representative single-shot image of OH from half of the duct is included in Fig. 28, illustrating the capability of PLIF imaging of complex flow structure.

Measurements were performed over a range of simulated flight conditions between Mach 4 and 6 [142,143]. Instantaneous images revealed the presence of large-scale structures, while averaged images at different axial locations showed the effects of total temperature and dynamic pressure on flame stability. Increasing temperature broadened the time-averaged flame zone, while increasing dynamic pressure tended to force the flame against the combustor sidewall. OH PLIF images were especially effective in evaluating changes in mixing associated with varying the fuel injection angles. Such measurements demonstrate the power of instantaneous and time-averaged 2-D PLIF for practical propulsion engineering. Details of these results and the experiments can be found in Refs. [142,143].

### 3.3. Unstart sensing in scramjets

Non-uniform temperature and gas composition along the line-of-sight (LOS) of a laser-absorption sensor can be evaluated using multiple laser wavelengths to simultaneously monitor

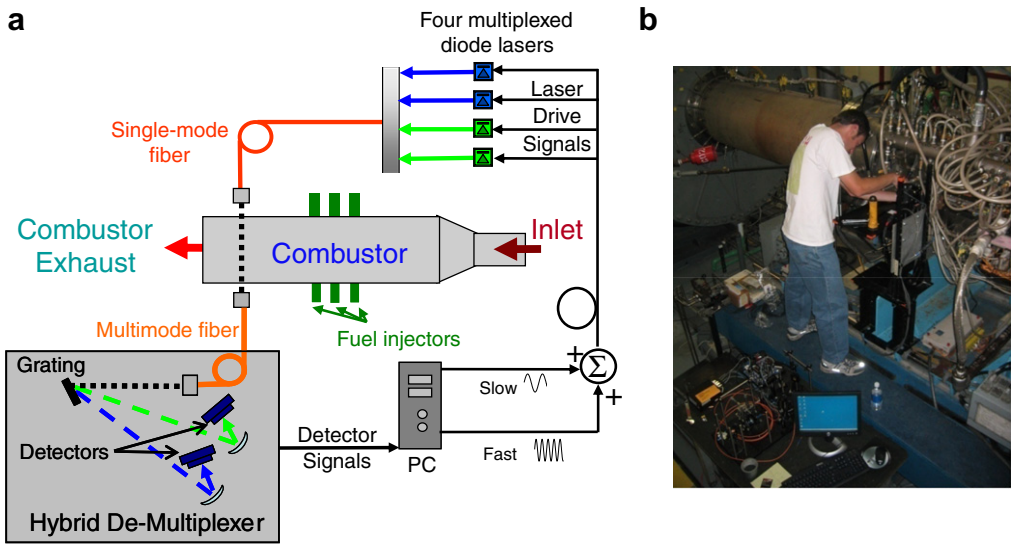


Fig. 29. (a) Schematic of the wavelength-multiplexed diode laser sensor used for measurements of temperature nonuniformity [112]. (b) Installation of the diode laser sensor in the scramjet test facility at AFRL, Wright Patterson AFB OH.

multiple transitions of the same species with different lower state energies $E''$. Use of such a sensor to discover incipient scramjet unstart provides an example of this promising detection strategy [112]. Absorption measurements were conducted in the model scamjet combustor test facility at AFRL (the same facility used for the PLIF example in Fig. 28, and shown in the photo on the right panel of Fig. 29). Here four diode lasers were multiplexed onto a single fiber and directed across the facility just downsteam of the combustor, as illustrated in the left panel of Fig. 29. Two line-pairs were selected, one with lower values of $E''$, more sensitive to cold regions of the flow ($T_{low}$ in Fig. 30), and the other with higher $E''$ values, more sensitive to hot regions of the flow ($T_{high}$ in Fig. 30). A hybrid combination of wavelength- and frequency- demultiplexing was used to separate the four laser signals in the transmitted light beam. The grating dispersed one pair of lasers (7444.4 and 7472.2 $cm^{-1}$) onto one detector and the second pair (7185.6 and 7205.2 $cm^{-1}$) onto the other detector. For each pair of lasers the signals were separated by frequency demultiplexing.

The measurement of temperature versus time $T_{low}(t)$ and $T_{high}(t)$ is shown in the left panel of

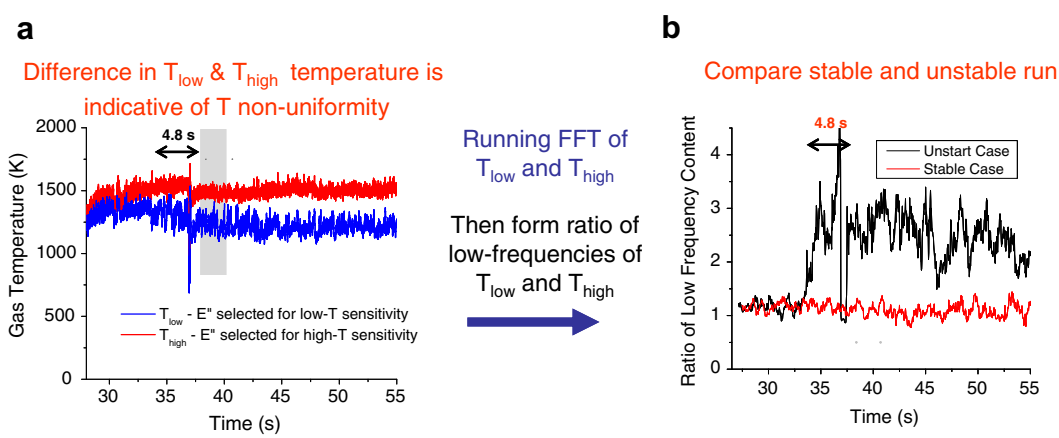


Fig. 30. (a) Temperature versus time for $T_{low}$ (blue) and $T_{high}$ (red) for a scramjet run with unstart; (b) Low-frequency fluctuations of temperature for an unstart run (black) and a stable run (red) (from Ref. [112]).

Fig. 30 for a scramjet run where unstart occurred as indicated by pressure sensors during the gray bar on the figure. If the temperature and gas composition were uniform along the line-of-sight the value for $T_{high}$ would equal $T_{low}$; the difference observed reflects the flow nonuniformity along the LOS. Although it is difficult to quantify this nonuniformity in temperature, fluctuations in the ratio of the two temperatures provides a simple measure of fluctuations in temperature nonuniformity. For example, if the boundary layer of the flow is colder (or hotter) than the core flow, fluctuations of the temperature nonuniformity (given by the ratio of the high- and low-$E''$ two-line temperatures) provides a useful measure of fluctuations in the boundary layer. Short-time Fourier transforms (FFT) of $T_{high}(t)/T_{low}(t)$ were used to monitor these fluctuations in temperature nonuniformity.

The low-frequency fluctuations from the ratio of $T_{high}(t)/T_{low}(t)$ are plotted in the right panel of Fig. 30 for two scramjet runs, one with stable operation and one that experienced unstart. The conditions were identical except that the unstart run had slightly more fuel added. In the unstart case the low-frequency fluctuations dramatically rose several seconds before the unstart event occurred. It is presently unclear what physical processes produced these fluctuations in the temperature nonuniformity, but because the increase in low-frequency fluctuations began in advance of unstart, one can see the potential utility of such measurements as a control variable.

The use of frequency or fluctuation analysis of line-of-sight absorption data is relatively new but is likely to find growing use in the future [112,144].

### 3.4. Investigations of pulse detonation engines

Absorption diagnostics are also ideal sensors for the non-intrusive evaluation of new propulsion concepts such as pulse detonation engines (PDEs). A number of PDE studies have been conducted with tunable diode laser absorption [145–149] yielding time-resolved measurements of temperature, pressure, velocity and species, in both laboratory and full-scale engine ground test. The success of laser absorption in such hostile and dynamic environments provides tangible evidence of the power and utility of quantitative laser-based absorption in propulsion devices. Reported here are two example studies, one utilizing an atomic cesium (Cs) tracer for thermometry and another providing real-time sensing of fuel loading.

Measurements of Cs tracer were used for simultaneous determination of time-resolved static temperature and pressure behind detonation waves in the single-cycle laboratory PDE illustrated in the top panel of Fig. 31 [147]. For these experiments 5 ppm of atomic cesium was seeded into the engine intake in the form of CsO dust. The static pressure was determined from the pressure-broadened absorption lineshape, which dramatically changed during the blowdown behind the detonation event as illustrated in the bottom panel of Fig. 31. The visible VCSEL (vertical-cavity surface-emitting laser) was pulsed at 500 kHz with a 1 μs "on-time". During each pulse the laser scanned approximately 1 nm in wavelength across the atomic Cs absorption near 852 nm for the transition from the $^2S_{1/2}$ ground state to the $^2P_{3/2}$ excited electronic state. The bottom panel of Fig. 31 shows three single sweeps of the laser wavelength. The first spectral sweep at ~606 μs after ignition, but before the detonation wave arrived at the sensor location, showed no absorption as the Cs remained in CsO dust; the overlay of an etalon trace on the first laser sweep was used to relate measurement time to laser wavelength. The second laser scan ~608 μs after ignition showed the pressure-broadened Cs absorption lineshape, and the third scan ~4.884 ms after ignition showed the narrowing of the Cs absorption lineshape as the pressure drops ~4 ms after the passage of the detonation wave. A Voigt lineshape analysis was used to include contributions from translational (kinetic) temperature (Doppler broadening) and pressure (collision broadening). The measured lineshapes were fit to determine both pressure versus time and kinetic temperature versus time (denoted as $T_{kin}$ in the temperature versus time plot in Fig. 32). Although not shown, the pressure versus time agreed well with independent measurements of time-resolved pressure using wall-mounted transducers.

Temperature could also be inferred from the ratio of the Cs population in the excited state to that in the ground state, assuming thermal equilibrium of the electronic state populations. The ground-state population was determined by optical absorption and the excited state by emission from thermally excited atoms; the ratio of emission to absorption provided a temperature as illustrated in Fig. 32 by the curve $T_{elec}$. Note that the two different measurements of temperature are in excellent agreement. The temperature measurements are compared in Fig. 32 with simulations performed at the Naval Research Laboratory (NRL), with good agreement within the combined uncertainty of the simulations and experiments [150].

The performance of PDEs, like all propulsion devices, depends on the efficient combustion of fuel. Infrared laser absorption measurements of fuel in the PDE at the US Naval Postgraduate School illustrate the use of this quantitative diagnostic to identify problems and solutions in fuel utilization [151]. The JP-10 fuel was monitored by absorption near 3.4 μm using wavelength-tunable mid-IR light produced by difference-frequency-generation (DFG) of two telecommunications

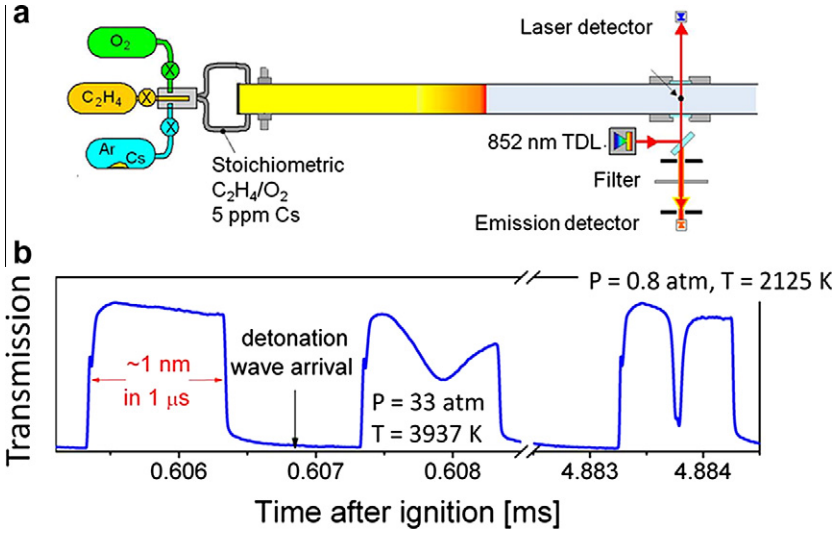


Fig. 31. (a) Single-cycle pulse detonation engine with laser absorption and optical emission sensors near the end of the detonation tube. (b) Three time-resolved wavelength scans of the diode laser sensor during the blowdown after detonation. The absorption lineshape provides time-resolved temperature and pressure (see text, from Ref. [147]).

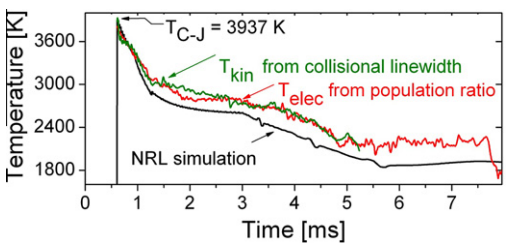


Fig. 32. Time-resolved temperature from the PDE in Fig. 31. Measured temperature compared with simulation calculations [150] (see text from Ref. [147]).

diode lasers in periodically poled lithium niobate (PPLN) [152]. As illustrated in the left panel of Fig. 33, the laser light was coupled into optical fiber, directed across the engine, collected into a second fiber, and directed to the detector, which allowed the lasers and detectors to be remotely located from the engine located in a separate room (test bay) owing to safety and noise considerations. Examples of fuel versus time data are shown in Fig. 33. In the unfired case, the pulse of unburned fuel passed the measurement station located slightly downstream of the fuel injectors with a pulsewidth of ~35 ms. In the fired case, the fuel rose until the flow is overtaken by the detonation wave; after the detonation wave passed, the fuel signal dropped to near zero showing complete or near-complete combustion. However, ~20 ms after the detonation wave, a packet of unburned fuel passed the detector. These measurements led to the discovery of crevices in the fuel loading plumbing where fuel was temporarily trapped, unable to burn, and did not contribute to thrust. Since the total fuel loading was known (needed to evaluate fuel-specific thrust), these packets of unburned fuel led to misinterpretation of the engine thrust versus fuel consumption. Elimination of these crevices improved PDE performance. This application of real-time *in situ* fuel sensing provides further tangible evidence of the utility of laser absorption measurements for propulsion research and development.

### 3.5. Control of swirl-stabilized gas turbine combustor

The need for improved fuel efficiency and more environmentally friendly emissions have motivated development of combustors that operate at leaner fuel/air equivalence ratios, where lower flame temperatures reduce the production of $NO_x$ and CO and increase lifetime of engine components. However, fuel-lean combustion is susceptible to instabilities in the form of thermoacoustic oscillations and lean blowout, which pose a serious problem for the operation of low-emission gas turbine combustors. Intensive experimental and theoretical work has been performed during the past decade to understand the driving mechanisms of thermoacoustic instabilities and to develop effective approaches to suppress these instabilities in laboratory-scale and full-scale combustors [153–157]. Because of the complex physical and chemical interactions involved, it is difficult to predict unstable combustion behavior, and typically combustion instabilities have been reduced or eliminated in industrial combustors by passive or active control mechanisms [155–157]. Here we provide an example of

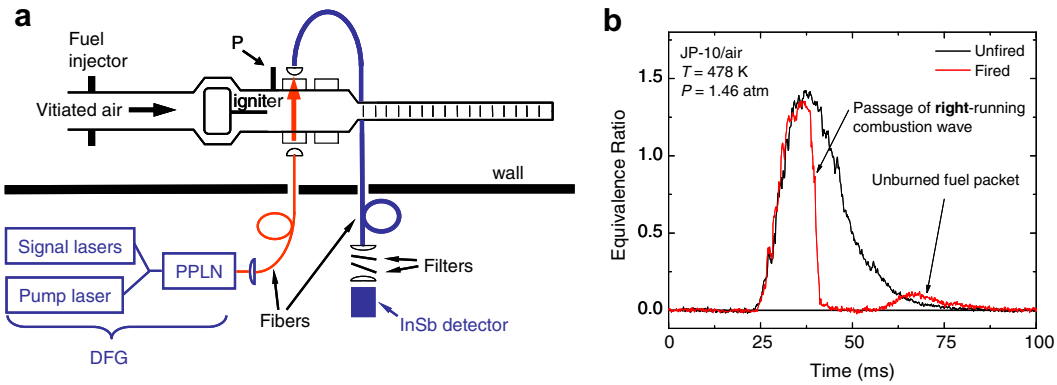


Fig. 33. (a) Schematic of the experiment showing the fiber-coupled mid-IR DRG laser source, PDE with fuel injection port and measurement station, and fiber-coupled detection of the transmitted light. (b) PDE fuel loading (JP-10) measured by laser absorption with the engine unfired and fired. When the combustion wave passes the detection laser beam in the fuel/air filled tube, fuel drops to near zero, but later arrival of unburned fuel revealed the presence of fuel trapping in the crevices of the valve system [152].

diode-laser absorption sensing for combustor control.

Lean blowout (LBO) occurs at low equivalence ratios or during rapid power transients, and is a major concern in the development of modern aeroengines. Currently, LBO is prevented by operating the combustor with a wide safety margin above the uncertain LBO equivalence ratio limit. This LBO limit varies with many operating parameters including the air and fuel flow rates, fuel composition, and combustor age [158, 159]. Consequently, pollutant emissions could be reduced and engine performance improved by operating with a narrower LBO safety margin. Previous work to investigate the mechanisms of LBO has found that the transition between stable flames and LBO is characterized by an intermediate stage with large-scale unsteadiness, local extinction and re-ignition events [158–162].

These observations of large-scale unsteadiness suggested the possibility of line-of-sight absorption sensing of temperature fluctuations as a potential control variable for LBO [144,163]. The left panel of Fig. 34 illustrates absorption sensing of temperature fluctuations in a swirl-stabilized flame chosen as a model gas turbine combustor at Stanford. A wavelength-modulated diode laser was scanned through two $H_2O$ absorption transitions near 1.4 μm at 2 kHz [164], and a running fast-Fourier transform (FFT) of the time-series of the ratio of absorption of these two transitions provided a fluctuation spectrum of the gas temperature. Even though the gas composition and temperature along the line-of-sight were not uniform, the fluctuations

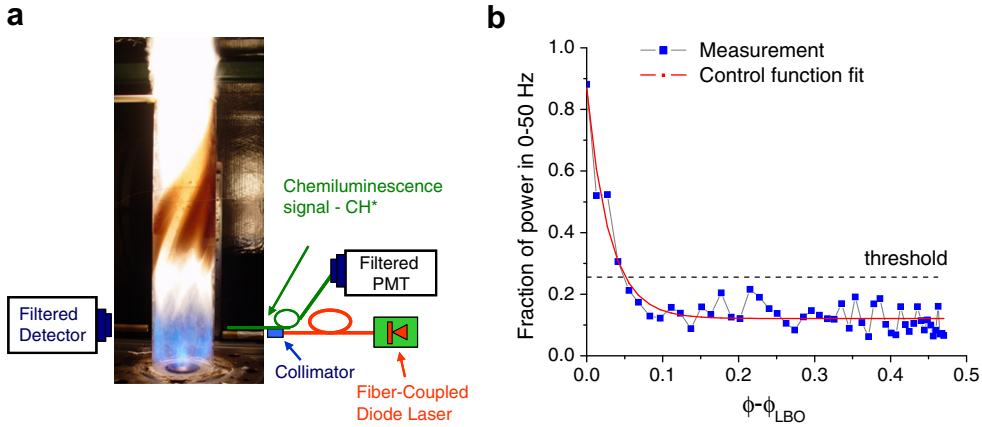


Fig. 34. (a) Swirl-stabilized propane/air flame with diode laser absorption sensor for temperature fluctuations. (b) Low-frequency fluctuations from running FFT of temperature vs. fuel–air stoichiometry near LBO [144,163].

observed provided a direct measure of flame stability. In particular, the low-frequency fluctuations from this FFT versus fuel–air stoichiometry showed a very pronounced increase as the flame approached LBO. Based on this signature, an active LBO control system was demonstrated without knowing the actual LBO limit, and the flame was controlled with a fuel–air stoichiometry within $\varphi - \varphi_{LBO} < 0.02$. This work illustrates the potential merit of absorption-based control sensing. Note that since the laser light is collimated to a small diameter beam, the sensor provides a localized measurement, free of the large backgrounds sometimes present on traditional pressure or chemiluminescent sensors. Future work to investigate this LBO sensor strategy for large-scale, high-pressure combustors is warranted.

### 3.6. Hypersonics test facility characterization

Diode laser absorption is an ideal diagnostic for the near-uniform line-of-sight conditions in well-designed, high-speed flow facilities, and offers the potential to accurately measure species, temperature, velocity and derived quantities such as mass and momentum flux, while minimizing disturbance to the test environment. Mass flux is an important parameter in air-breathing propulsion, and is used in calculations of inlet and combustor performance; exit momentum flux is related to thrust and hence is also a critical performance parameter. Measurement of mass flux can aid in propulsion test facility operation and provide test data needed for accurately evaluating engine performance. Diode laser absorption sensing of mass flux was first explored by Philippe and Hanson [94,95], patented for thrust measurements in 1993 [96], and used for practical ground test measurements in a commercial turbofan (PW6000) inlet at Pratt and Whitney [99], and a full-scale Pratt & Whitney F-100 engine in a ground test stand at the NASA Dryden Flight Research Facility [97].

As illustrated in Fig. 35, to determine velocity the laser beams are directed upstream and downstream in the flow and the laser wavelength is scanned to capture the differential Doppler shift between the absorption lineshapes as illustrated on the right side of Fig. 35. The mass flux is the product of velocity and gas density, so a second laser is added to determine gas temperature by two-line thermometry. The density is then found from the ideal gas equation of state and knowledge of pressure. The concentration of the absorbing species may be determined from either of the absorption signals. We recently constructed a sensor of this design to validate the flow conditions in the direct-connect supersonic combustion test facility at NASA Langley. In this facility, the gas is vitiated (heated by combustion of hydrogen with oxygen replenishment). Hence the sensor used $H_2O$ absorption to exploit mature telecommunications diode lasers in the 1.4 μm region, providing access to the $2\nu_1$, $2\nu_3$, and $\nu_1 + \nu_3$ absorption bands. The sensor also included recent developments in calibration-free WMS-2f temperature sensing [109,110], allowing simultaneous temperature and velocity measurements. Results are shown Fig. 36; validation data from a low-speed wind tunnel at Stanford are shown in the left panel, while data from the test facility at NASA Langley appear in the right panel. Note the excellent agreement with expected values over the very wide range in measured velocity. Details of these measurement campaigns were described recently in Ref. [100].

Expansion of gas heated and compressed by reflected shocks in a shock tunnel has been used for more than 50 years to produce high-enthalpy flows [165], yet the test conditions can have large uncertainties. The freestream conditions in these test facilities are calculated from non-equilibrium multi-dimensional nozzle codes and rely on measured shock speeds and side-wall pressures. Prior to measurements with tunable-laser absorption spectroscopy, there was a paucity of direct measurements of freestream conditions. More than a decade ago, optical fiber technology was exploited to develop the freestream probe illustrated in Fig. 37. The Stanford-built probe was deployed for measurements in the test section of the CUBRC (Research Center at the State University of New York, Buffalo) 96-inch hypersonic shock tunnel [166–168]. The first measurements used a large 18 cm probe span to simultaneously measure time-resolved temperature, velocity, and water vapor content in the freestream flow. An example of results is shown in the right panel of Fig. 37; these unique data were the first spectrally and spatially resolved absorption measurements in the freestream of a hypersonic shock tunnel. A second-generation probe with only a 5 cm gap was then assembled and tested successfully at CUBRC. Subsequently, an even smaller probe with a 2 cm span was successfully employed to measure velocity via the low levels of naturally present potassium vapor. Note that the nozzle exit in this facility is 121 cm, so all of these probes were able to sample the freestream with good spatial and temporal resolution.

Tunable diode laser absorption sensors have also been exploited to monitor O- and N-atom concentrations in the 60 MW arc heater which drives hypersonic test facilities at NASA Ames Research Center [169]. These arc-heated tunnels are used for a wide range of high-enthalpy testing including evaluation of the thermal protective barriers for reentry vehicles, e.g., space shuttle tiles. The hypersonic flow properties produced by nozzle expansion are typically calculated using non-equilibrium, multi-dimensional, nozzle codes with energy-balance considerations, ideally including knowledge of the stagnation tempera-

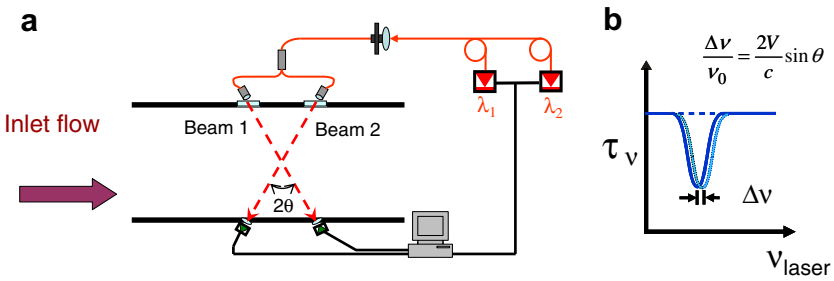


Fig. 35. (a) Schematic of a laser-absorption sensor to determine velocity via the Doppler shift of the two absorption lineshapes; (b) illustration of shifted lineshapes.

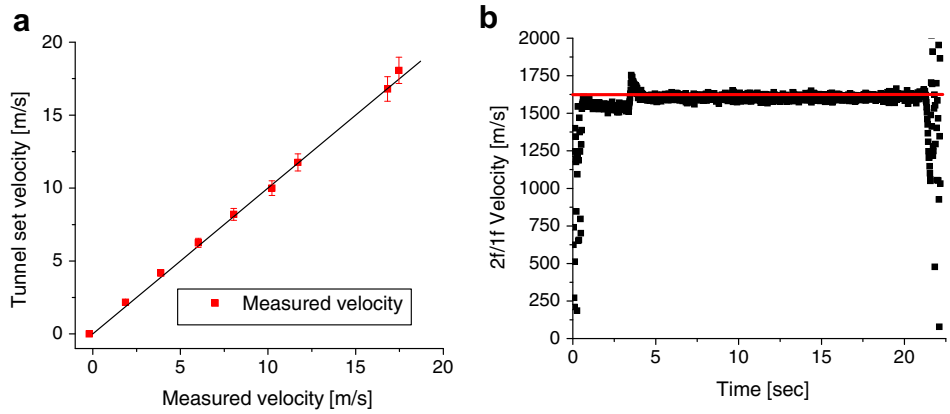


Fig. 36. (a) Velocity data in low-speed tunnel at Stanford and hypersonic combustion test facility at NASA Langley from Ref. [100].

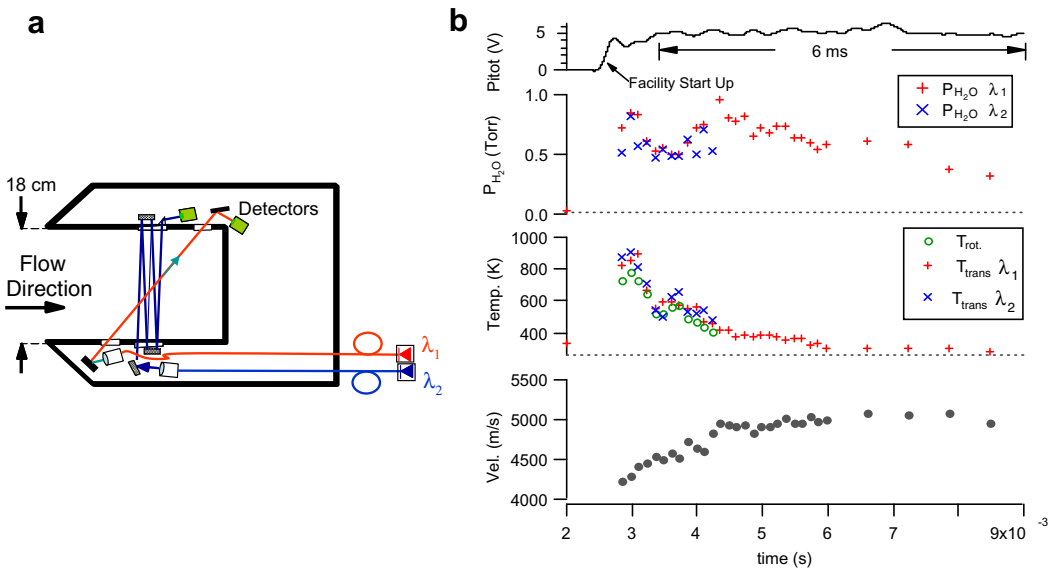


Fig. 37. (a) Fiber-coupled hypersonic flow probe for velocity and temperature using $H_2O$ absorption; (b) example measurements of time-resolved (5 kHz) pressure, velocity and temperature measured at CUBRC (taken from Ref. [168]).

ture of the arc-heated gas. At the very high temperatures (5000–9000 K) and multi-atmosphere pressures in the arc heater, air is dissociated and in thermodynamic equilibrium so that the temperature can be inferred from the concentration of atomic O or atomic N in electronically excited states. Even at these high temperatures, the excited-state populations are small, less than 100 ppb, because these states lie far above the ground electronic state (>9 eV). However, absorption spectroscopy with tunable diode lasers was successfully used to measure O- and N-atom concentration and temperature with good sensitivity and accuracy [169]. Time-resolved *in situ* absorption measurements of atomic copper were also demonstrated as a real-time method for monitoring erosion from the arc-heater electrodes.

Lastly, we note the successful use of diode laser absorption sensors for validating conditions in shock tubes at Stanford, as needed for chemical kinetics measurements described in the first section of this paper. Especially when the chemical reactions release energy (oxidation reactions) or consume energy (pyrolysis reactions) independent measurements of gas temperature validate critical aspects of shock tube kinetics modeling. Recently we have used absorption of $CO_2$ in the bands near 2.7 μm and WMS-2f/1f techniques to determine temperatures behind reflected shocks with precision of better than ±3 K at 1000 K, as discussed earlier in this paper [18].

### *3.7. Future of laser-based diagnostics for propulsion studies*

#### *3.7.1. Accomplishments*

Laser-absorption sensing for propulsion applications has now progressed well beyond initial proof-of-concept demonstrations. For example, PLIF imaging has become a routine research tool for the characterization of flowfield structure for a wide variety of propulsion and engine applications. Similarly, laser absorption sensors have been used successfully to monitor conditions in ground test facilities for more than a decade. On the other hand, absorption sensors are just beginning to emerge for combustion control, engine characterization, and for *in situ* measurements in aeroengine and airframe models undergoing tests in high-speed flow facilities. As the latter applications mature, fiber-coupled diode laser absorption sensors can be anticipated for in-flight applications.

#### *3.7.2. Needs and opportunities*

New frontiers for laser-based propulsion diagnostics include high-speed PLIF imaging, high-pressure combustion applications, the development of miniature fiber-coupled absorption sensors for flight applications, and the coupling of CW laser absorption with frequency analysis. Significant research progress has already been reported, some of which was referenced above, but much work remains to be done. High-speed imaging of flowfield structures holds the promise to validate time-resolved simulation codes for hypersonic reacting flows. Novel new low-power, fiber-based sensor arrays may bring tomographic imaging inside aeroengine and airframe modes for hypersonic ground test. CW laser absorption with fast frequency analysis may provide a new diagnostic strategy for achieving spatial resolution along a line-of-sight, or for studying (or monitoring) jet noise. Such sensor systems also have the potential for new paradigms of combustion control.

## 4. Practical combustion systems

Laser diagnostics measurements have contributed much to our understanding of a multitude of practical combustors, and the development of laser-based techniques for such applications has often been discussed. In the introduction, two reviews were highlighted: (1) the Hottel plenary of the 27th combustion symposium [1] and (2) the 50th anniversary plenary on laser diagnostics at the 30th symposium [2]. In this section, we illustrate the maturity of some laser absorption diagnostics that have begun to transition from research devices to "sensors" for *in situ* monitoring of a wide variety of practical combustion systems and, in some cases, to provide signals for closed-loop control to solve important combustion problems. The distinction between laser absorption *diagnostics* and *sensors* is defined. Then we turn to the use of these sensors in practical combustion systems where the ultimate goal is control, to optimize combustion efficiency, minimize combustion emissions, and provide for safe combustor operation. The combustion researcher typically selects the diagnostic method to provide the best data possible to test a hypothesis, validate a model (or simulation code) or to develop qualitative understanding. In contrast, the selection of a sensor for practical combustion systems is primarily driven by a cost/benefit analysis of the installation, operation and maintenance costs versus the monetary value of the improved performance. These sensors often must operate 24 × 7 (24 hours a day, 7 days a week) with minimal attention by operators without technical backgrounds. Thus, the transition of a laser-absorption diagnostic to a practical commercial sensor requires significant engineering efforts beyond proof-of-principle demonstration measurements. Here selected benchmark demonstrations are highlighted, which have the potential to survive the engineering development and cost/benefit analysis and transition into practical sensors with eventual wide use by industry. We conclude with a short discussion of sensor commercialization

choosing as an example the laser-absorption sensors for ammonia that are now commercially available for combustion applications. These sensors are being used to minimize/control excess ammonia (ammonia slip [170]) in the reduction of $NO_x$ emissions from coal-fired power plants using either selective catalytic reduction (SNR) or selective non-catalytic reduction (SNCR) technologies [171].

### 4.1. Sensor fundamentals

In addition to the physical principles of laser absorption, discussed previously in this paper, practical sensors must also consider the availability, cost and maintenance needs of robust lasers, fiber optics, access windows, and detectors. These practical parameters vary widely with laser wavelength, and Fig. 38 shows a current assessment of diode-laser-based technology versus wavelength. There has been significant investment by the optical storage, optical reader, and telecommunications industries in the visible and near-infrared region from 0.6 to 1.75 μm, and the laser technology is relatively mature with readily available inexpensive lasers that have been performance tested over a wide range of environmental conditions and shown to be reliable for continuous operation beyond one year. At longer wavelengths, diode lasers are now available from 1.75 to 2.9 μm, although these devices do not yet have a history of long-term performance testing over a wide range of environmental conditions. In the mid-infrared (3–10 μm), significant progress has recently been made in the development of laser sources. For example, in the region from 3 to 4.4 μm, a robust source of laser light has been developed using DFG of two robust telecommunications devices [172]; also in this same wavelength region, interband cascade lasers (ICL) are beginning to emerge [173]. Beyond 4 μm, pulsed and CW quantum cascade (QC) lasers have recently become available [174]. These emerging lasers (DFG and QC) have been demonstrated to perform continuously for months as research sensors to monitor trace gases in the atmosphere [175]. In practical combustion systems, the local environment can be quite hostile, e.g., the ambient temperature in a coal-fired power plant near the boiler can be in excess of 50 °C with wall temperatures in excess of this value. Therefore, it is very desirable to fiber-couple remotely housed lasers to the combustion rig. This fiber-coupling provides the visible and telecommunications regions of the spectrum a strong advantage, as in this region the fiber optics technology is well developed and a wide variety of components have off-the-shelf availability. However, more expensive and less-robust optical fibers are becoming available for wavelengths up to 4 μm.

At the elevated temperatures of combustion systems, water vapor has at least weak absorption over the entire spectral region from 1–6 μm as illustrated in Fig. 39 for 700 K, and the gaps in the $H_2O$ spectrum fill as temperature is increased. (Note that the temperature dependence of the $H_2O$ absorption coefficient is illustrated by comparison of Fig. 39 with the 1500 K $H_2O$ absorption spectrum in Fig. 26.) In post-combustion gases the water vapor mole fraction is large (~10%), so even weak absorption transitions of $H_2O$ can produce major interference for detection of trace combustion products (here NO and CO are used as examples) even when using the strongest transitions of the trace species. The absorption bands of the small molecules illustrated in Fig. 39 are composed of discrete transitions, and sensitive detection of trace species generally relies on selection of transitions between those of $H_2O$, as illustrated in Fig. 40 for CO line selection in the 2.3 μm band. Ten candidate CO lines are identified (by arrows) as promising in terms of avoiding excessive $H_2O$ interference. Because the absorption spectra are very temperature sensitive, the line selection must be investigated for the specific temperature range where the sensor will be used in the combustion system.

The species NO, CO, and $NH_3$ are excellent candidates for combustion control applications in large-scale commercial combustors, and examples of tunable diode laser sensing in practical systems of these molecules will be discussed below. Because of safety considerations, many practical combustion systems have excess oxygen in the

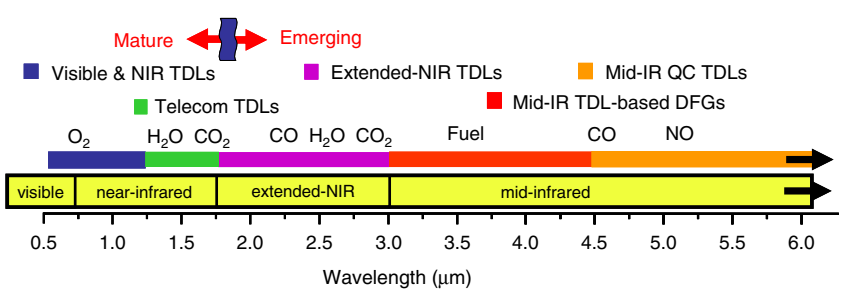


Fig. 38. Diode laser types and example combustion species versus wavelength.

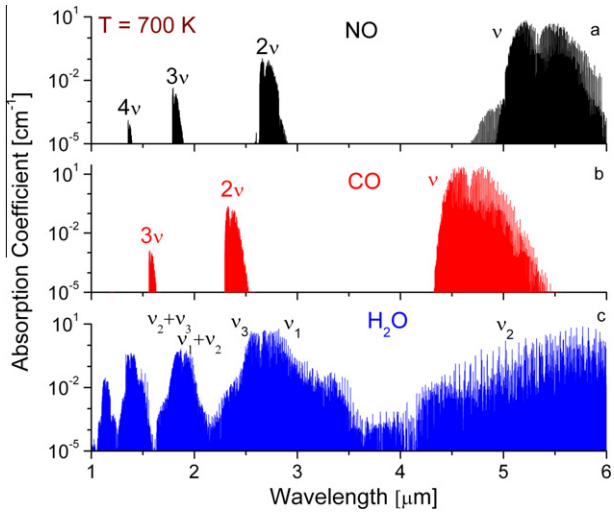


Fig. 39. Absorption coefficient at 700 K for NO in panel (a), CO in panel (b) and $H_2O$ in panel (c) as calculated from HITRAN [36].

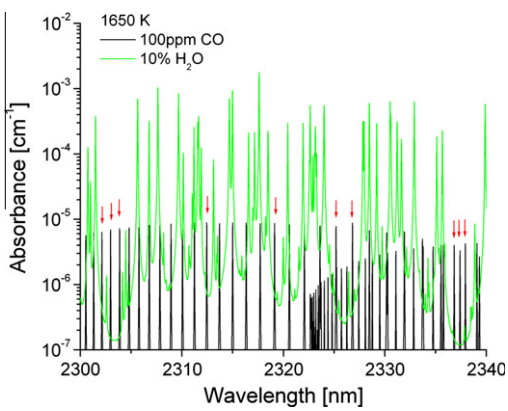


Fig. 40. Absorbance per cm pathlength for 10% $H_2O$ and 100 ppm of CO at 1650 K and atmospheric pressure in the region of the first overtone band of CO as calculated from HITRAN [36].

exhaust. However, fuel efficiency is optimized for stoichiometric combustion. Therefore, sensors that would safely allow minimization of excess oxygen provide financial incentive for improved combustion control. However, there are also other financial incentives to reduce combustor emissions. For example, it is generally more cost effective to reduce the production of pollutant emissions than it is to scrub them from the exhaust stack. Referring to control systems such as SNR or SNCR of NO, the cost of the urea or ammonia reagents can be reduced by adding only the minimal amount required for the NO removal in the exhaust gases; if too much reagent is added, the cost increase also includes the need to scrub the excess $NH_3$ from the exhaust. In the following, the use of diode laser absorption sensors will be discussed for three applications: (1) incineration, (2) power plant combustors, and (3) the exhaust stream of power plants. All of these sensors have significant promise for commercial applications although their current status spans the development curve from proof-of-concept research to commercially available sensors.

### 4.2. Waste incinerator control

Research in support of compact waste incinerators for US Navy ships [176–178] led to the first demonstration of laser-based combustion control by Furlong et al. [179–181] In this application as illustrated in Fig. 41, the solid waste is converted to gaseous products using a starved-air pyrolysis chamber and the hazardous components are incinerated in a secondary oxidation chamber or afterburner. These incinerators were designed for retrofit of existing systems that did not meet increased regulatory scrutiny on emissions. Therefore the physical space available required a compact and efficient design that led to research conducted on 5 and 50 kW model afterburners. Acoustic forcing was selected as a means of promoting efficient mixing of the pyrolysis products with the air and fuel in the afterburner, and both the afterburner air and the afterburner fuel were modulated. A combination of PLIF of OH, chemiluminescent emission of OH and total flame luminosity were used first to visualize the changes in the flame structure with the relative phase between the two modulated flows, as illustrated in Fig. 42. These advanced laser diagnostics (i.e., optical imaging) provided the understanding of flame structure needed to design proper placement

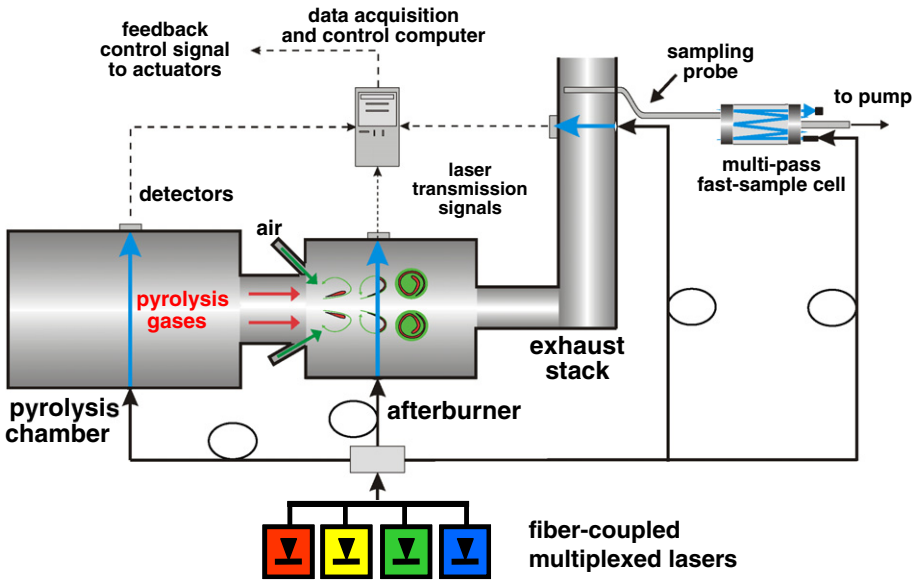


Fig. 41. Concept drawing for shipboard incinerator where the hazardous gas stream from waste pyrolysis is burned in a acoustically forced afterburner. Two diode lasers monitor temperature in the afterburner (and pyrolysis chamber), while others monitor unburned hydrocarbons, $CO_2$ and CO sampled from the exhaust into a multipass cell (from Refs. [176–178]).

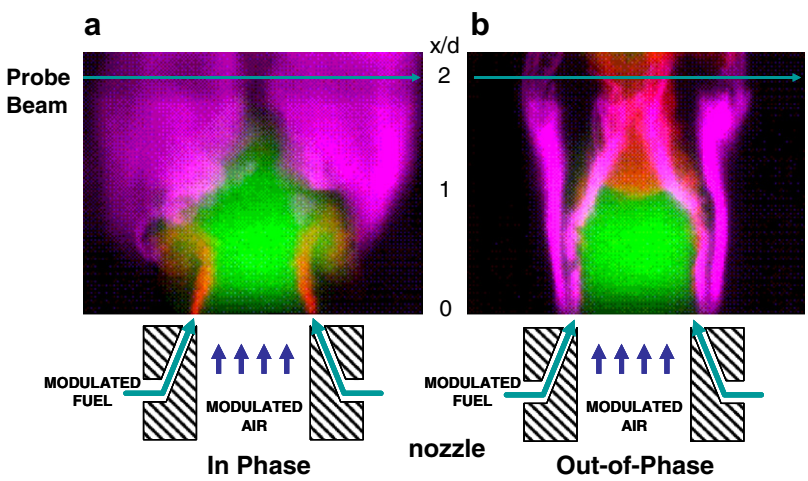


Fig. 42. Flame structure of the incinerator afterburner for modulation of the input air and fuel (a) in-phase and (b) out-of-phase as determined by a combination of OH PLIF, chemiluminescent emission of OH, and flame luminosity (from Ref. [176]).

of an absorption line-of-sight to optimally capture temperature fluctuations driven by the acoustic forcing. The simple control concept was that the most efficient (complete) combustion would occur with the greatest temperature excursions during the passage of well-mixed large-scale combusting structures. In other words, the goal of the control system was to seek the highest temperature excursions associated with intermittent passage of hot (combusting) and cold structures.

The temperature sensor consisted of two diode lasers emitting near 1.34 and 1.39 μm, selected to monitor two absorption features of $H_2O$ with different lower-state internal energy ($E''$). The light from these two lasers was fiber-combined and directed though the afterburner. The transmitted light was dispersed on a diffraction grating and the light from each laser was collected onto a separate detector. The two lasers were scanned at a 10 kHz rate providing integrated absorbance mea-

surements on the two transitions every 100 μs. This ratio showed large periodic oscillations at the forcing frequency ($f_0 = 100$ Hz) that were suggestive of strong, coherent vortices, and the strength of these oscillations depended on the relative phase of the fuel and air forcing. Although the ratio of the absorption of two transitions could provide the absolute average temperature along the line-of-sight, in this application the temperature fluctuation was a more suitable target. A FFT of a time-series of the ratio of the two absorbance values showed a strong fluctuation signal at the modulation frequency, and as observed in the time series data, the strength of the fluctuations depended on the relative phase of the fuel and air forcing. The $T_{rms}$ at the forcing frequency determined from the FFT of the absorbance ratio proved to be an excellent control variable to optimize the relative phase between the fuel and air forcing and therefore achieve the highest combustion temperatures. These highest temperatures corresponded to the highest combustion efficiency of the waste material, as validated by measurements of CO and unburned hydrocarbons in the combustion exhaust [180, 181]. The use of temperature fluctuations as a control variable mitigated the uncertainty in absolute temperature due to non-uniformities in temperature and $H_2O$ concentration along the line-of-sight.

The efficiency of the afterburner was determined by the ratio of CO and $CO_2$ in the exhaust as measured by extractive sampling into a multi-pass cell. The extracted gas sample was dried and filtered to remove particulate. The long 30 m optical path in the fast-flow multi-pass cell provided the opportunity to use telecommunications wavelengths to monitor weak transitions of CO and $CO_2$ in cold (300 K) combustion exhaust. Figure 43 shows the CO measured in the 5 kW model afterburner at a normalized distance downstream ($x/D$, $D$ = diameter of the afterburner) equal to 14 and the $T_{rms}$ at $x/D = 4$. The CO was minimized at the maximum $T_{rms}$; similarly, the unburned hydrocarbons were also minimized when $T_{rms}$ at the forcing frequency was maximized. $T_{rms}$ was used as a control variable for a variety of control algorithms and optimum closed-loop control of the relative phase was attained in times as short as 30 control steps [180]. The amount of CO in the combustion exhaust was reduced by two-orders of magnitude by the control system as illustrated in Fig. 43b. The time to achieve control could be further reduced by at least 100-fold with modern electronics. When this sensor was then applied to a 50 kW model incinerator, the closed loop control of $T_{rms}$ dropped the exhaust CO from 2500 ppm to less than 200 ppm [181].

This pioneering demonstration of laser-based control, utilizing robust tunable diode lasers, illustrates a successful transition from research to a practical application. Similar sensors should be effective in optimizing combustion efficiency and reducing exhaust pollutants for other types of combustors.

Although the model shipboard incinerator application was the first closed-control demonstration with diode laser sensors, laser absorption to monitor ammonia slip for SNCR had been demonstrated in a large-scale municipal-waste incinerator control nearly a decade earlier by Wolfrum and colleagues [170,182]. SNCR and SNR reduces $NO_x$ by removing NO from the exhaust system by adding a reagent [171] (called Thermal de$NO_x$ for ammonia reagent [183]. RAPRENOx for isocyanic acid [184], and NOxOUT for urea [185]). The urea process is often preferred to other reagents [186]. The efficiency of the urea addition for SNCR requires that the injection into the exhaust stream occur in the narrow and fuel-dependent window where the exhaust gas temperature ranges between 1175 and 1300 K (the SNR temperature window is lower). If the exhaust gas temperature is incorrect or if too much reactant is added, ammonia escapes into the exhaust stream as an additional pollutant,

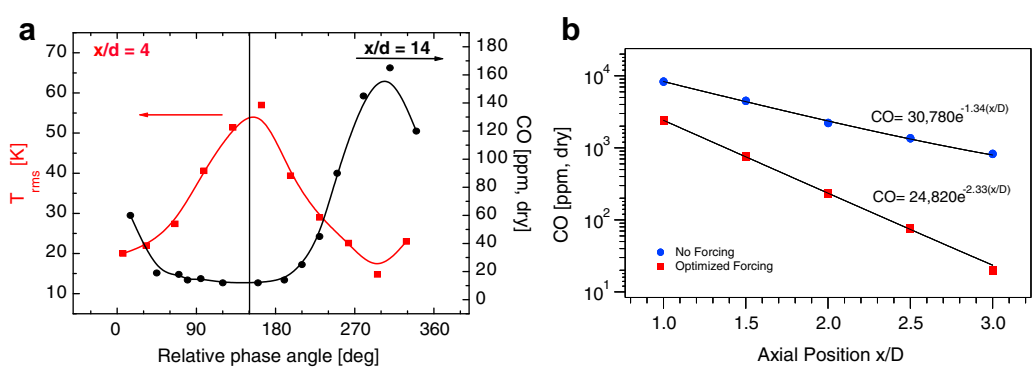


Fig. 43. (a) $T_{rms}$ and exhaust CO concentration measured by diode laser absorption sensors in a 5 kW model afterburner; (b) reduction in CO vs height in the burner with and without control; from Ref. [181].

which is termed ammonia slip [171]. The fugitive $NH_3$ from SNCR by urea was monitored in a 20 MW municipal waste incinerator using a $^{13}CO_2$ laser that was rapidly alternated between two emission lines (one of which is absorbed by $NH_3$). The sensor, called LISA (laser *in situ* ammonia sensor) has been tested in a variety of large-scale combustors in Germany [182,187].

Proof-of-concept diode-laser-absorption measurements of water vapor and oxygen were also performed in hot exhaust gases immediately after the combustor in the same 20 MW waste incinerator by Ebert and coworkers [188]. Significant challenges to these *in situ* measurements in large-scale facilities included the long optical paths, large emission backgrounds, transmission losses to particulate scattering, and the large mechanical motion of the combustor and exhaust ducts with combustor load. The absorption measurements of oxygen decreased with the increase in combustion product water vapor; however, the large fluctuations in both oxygen and $H_2O$ on multi-minute time scales illustrate the difficulties in achieving efficient combustion with the temporal variations in heating value of the fuel stream typical of a municipal waste incinerator. These variations in loading are exacerbated for batch-loaded rotary kiln incineration often used for hazardous waste. Incomplete destruction of hazardous waste is a serious issue, and absorption measurements of CO and oxygen were performed immediately after the kiln exit to suppress large bursts of CO in the exhaust and presumably improve the average combustion efficiency [189,190].

### 4.3. Power plant combustor characterization

*In situ* measurements in the combustor of a power plant provide the opportunity to identify maladjustments of individual burners in the multi-burner applications. The burner tuning can vary with the heating value of the fuel (especially a problem for combustors burning coal, coal gasifier products, or refinery gas), with variations in the input air, or with a malfunction in the burner or the fuel/air flow to the burner. Burner malfunctions are especially troublesome for gas-fired power plants where their rapid response can be used for load leveling demand on the power grid. In the event of an ignition failure, the build-up of unburned natural gas can lead to explosive consequences at the cost of tens of millions of dollars [186,191].

A diode-laser absorption sensor was demonstrated in a 1 GW gas-fired power plant by Ebert et al. that could identify burner ignition failure as well as burner tuning and general overall combustor health [189]. This large-scale plant had an internal cross section of $10 \times 16.5$ m with twenty individual burners arranged over five levels (total vertical distribution was ~25 m). Diode lasers for oxygen (near 762 nm), water vapor (near 811 nm), and methane and $CO_2$ (near 1.65 μm) were multiplexed onto a common optical path and directed across the combustor in the plane of the fourth burner level. These simultaneous measurements of $O_2$, $H_2O$, $CH_4$, and $CO_2$ were the first diode laser absorption measurements in the combustor of a large-scale power plant. Multiple engineering challenges were solved including the need for closed-loop control of the laser-beam alignment to account for the mechanical changes in the combustor walls (and mounting hardware) with the thermal load. Gas-fired combustion is relatively clean, but the transmission losses over the long path length were large (exceeding 95%) presumably due to beam steering by the thermal (and density) gradients. However, unburned methane could readily be detected with a 100 ppm detection limit and a two order-of-magnitude dynamic range [189]. Temperature could be monitored and ranged from 300 K before ignition to ~1300 K with all burners firing. The methane detection limit was more than adequate to provide the safety indication needed during ignition and to identify that one or more burners had failed to ignite.

The transition from gas-fired to coal-fired power plants significantly increases the degree-of-difficulty of laser-absorption sensing. The first measurements, again by Ebert and coworkers [192], were performed in a full-scale (600 MW) coal-fired power plant; see Fig. 44. Measurements of CO (near 1.65 μm), $H_2O$ (near 811 nm) and gas temperature from the ratio of two $H_2O$ absorption features were made at a height of 65 m above the floor of the $20 \times 20$ m combustion chamber. The gas temperature at the measurement location was ~1450 K where the CO transitions in the second overtone band all have significant interference from $H_2O$. However, the CO concentrations were high (>1% by volume) in this fuel-rich combustor, and the $H_2O$ measurements near 811 nm were used to correct the CO data for this interference. The laser light was highly obscured by optical scattering with transmission ranging from 0.02–0.04%. In addition, the luminosity of the of the hot particulate laden gas was as much as ten times the detected laser signal even with the narrowband 10 nm optical filters. However, good measurement sensitivity was possible by averaging data over a 30 s window. The laser-based measurement time was still roughly ten times faster than the gas sampling measurements, which quench-cooled the gases, filtered the particulate and then transported the sample more than 100 m for an NDIR measurement. These measurements provide further evidence of the potential for wavelength-scanned diode laser measurements in extreme environments.

Tunable diode laser sensors have been applied recently in large-scale, coal-fired power plants with use of tomographic methods to identify variations

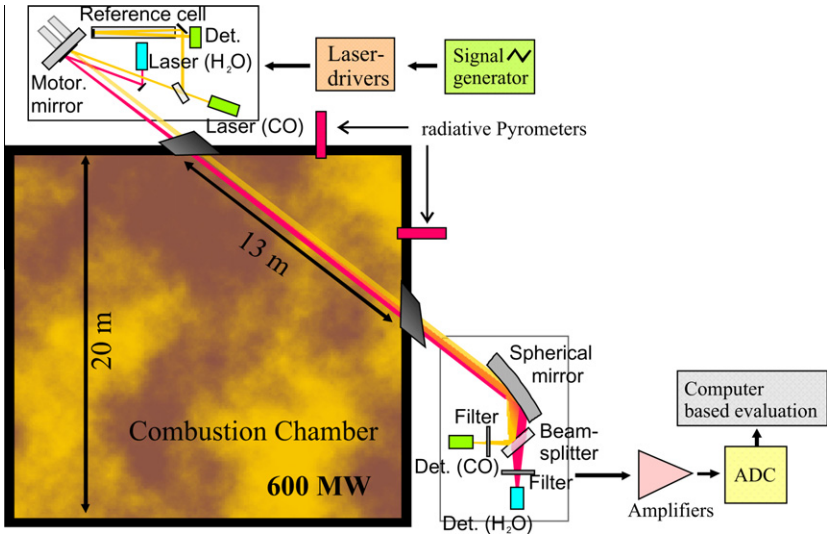


Fig. 44. Measurement arrangement for CO and $H_2O$ measurements in a 600 MW coal-fired power plant, 65 m from the combustor floor (from Ref. [192]).

in individual burner performance. In such combustors, when the CO concentration becomes large, the combustion environment becomes highly reducing, and the formation of coal slag can become an important problem. In catastrophic events, the slag can form into a mass that requires combustor shutdown. The facility downtime costs can exceed \$1 M/day when the utility demand is for 100% output, in addition to any capital costs of repair. Therefore, a sensor for simultaneous measurements of CO, $O_2$, and temperature was investigated by Sappey and coworkers for application to slagging and non-slagging conditions [193].

Simultaneous measurements of CO (near 1.65 μm), $H_2O$ (near 1.4 μm), $O_2$ (near 760 nm) and temperature measurements were made in a 500 MW coal-fired boiler. The CO data was corrected for $H_2O$ interference, similar to the CO measurements in a German coal-fired plant [192]. The temperature was determined by a fit to multiple water vapor absorption values. The measurements were performed along a rectangular grid of $5 \times 5$ equally spaced line-of-sight in the $16 \times 18$ m combustor, and the height of the measurement plane was about halfway between the combustor floor and the bullnose. Figure 45 shows the tomographic reconstruction of the absorption results. The top row corresponds to an optimized combustion condition and the bottom row corresponds to a slagging condition.

The CO concentration peaks at about 0.5% when the burners are properly tuned, but in the slagging case, the CO concentration peaks near the center of the burner at a very high value of 3.5% indicative of potential slag formation conditions. Note that individual measurements on a single line-of-sight have only a weak indication of slag formation conditions in the middle of the burner, but the spatially coarse reconstruction of only a $5 \times 5$ grid is sufficient to clearly identify the adverse condition. When the burner begins to slag the temperature drops and becomes more uniform as shown in Fig. 45, but the measured water vapor concentration is quite low near the wall suggesting reduced combustion efficiency near the side walls. Although absorption tomography sensors for power plants are in their infancy, this pioneering work reveals excellent potential for real-time burner optimization accounting for changes in the heating value of the input coal (e.g., coal containing a much larger dirt fraction) or changes in environmental conditions (e.g., wet coal being fed in a strong rain storm).

### 4.4. Power plant exhaust monitoring

There is a large research literature on the use of laser absorption for monitoring trace gases in ambient air or for compliance monitoring at the exit of the exhaust stack of large-scale combustors. Here we focus attention on the *in situ* detection and monitoring of trace pollutants (NO and $NH_3$) in the combustion exhaust gases at elevated temperatures prior to particulate clean-up and prior to low-temperature exhaust after-treatments.

NO and its oxidation product $NO_2$ are combustion-generated pollutants that contribute to ground-level ozone, acid rain and a wide variety of adverse human health effects. These issues have led to increasingly stringent regulations, e.g. for $NO_x$ emissions, nearly worldwide. Monitoring $NO_x$ (the sum of NO and $NO_2$) at the exhaust stack is mandated in the US and Europe and typically performed by chemiluminescent analysis of

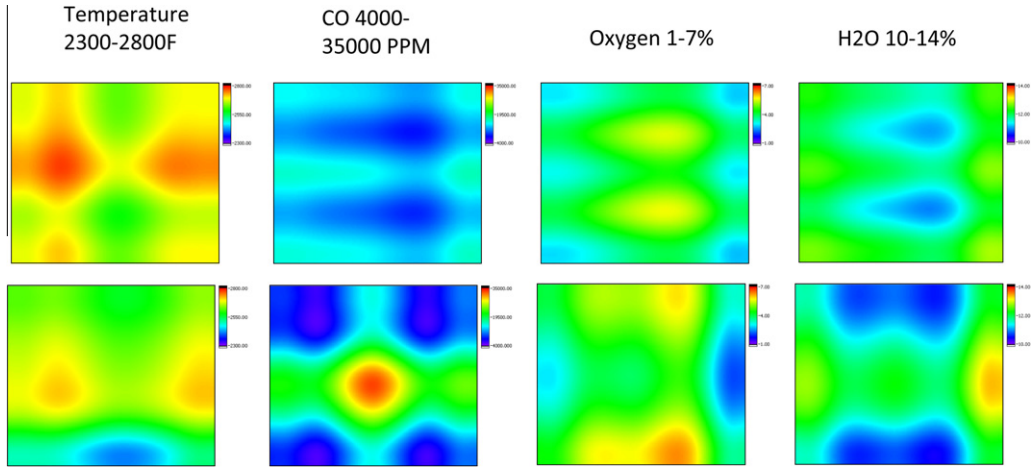


Fig. 45. Tomographic reconstruction of a 5 × 5 grid of laser absorption measurements in a 16 × 18 m coal-fired boiler combustor. The numbers indicate the full scale range of the spectral color scale (red represents high values). The top row corresponds to an optimized firing condition and the bottom row corresponds to a slagging condition (courtesy of Ref. [193]).

sampled gases. Accurate and reliable real-time sensors are an important part of any control system, and measurements from a variety of locations in the power plant, ranging from the combustor to the atmospheric exit of the exhaust stack, could facilitate the needed $NO_x$ control. The residence times in such power plants can be large, and thus *in situ* monitoring of NO closer to the combustion zone, especially just past any active $NO_x$ control systems (e.g., SNCR or reburn), has the potential for significantly more efficient $NO_x$ control. Very recently (December 2009), NO was monitored by laser absorption at the economizer exit of a pulverized-coal-fired boiler (270 MW) providing steam for the generation of electricity; the sensor location is shown in Fig. 46. This proof-of-concept measurement was conducted to determine if absorption of fundamental band transitions of NO near 5.2 μm could be used to monitor NO in the dust-laden exhaust of the large-scale, coal-fired boiler [194].

The fundamental-band transitions ($\Delta v = 1$) of NO are roughly 100 times stronger than the first-overtone transitions near 2.65 μm ($\Delta v = 2$) and 3000 times stronger than those of the second overtone near 1.8 μm ($\Delta v = 3$); see Fig. 39. Laser

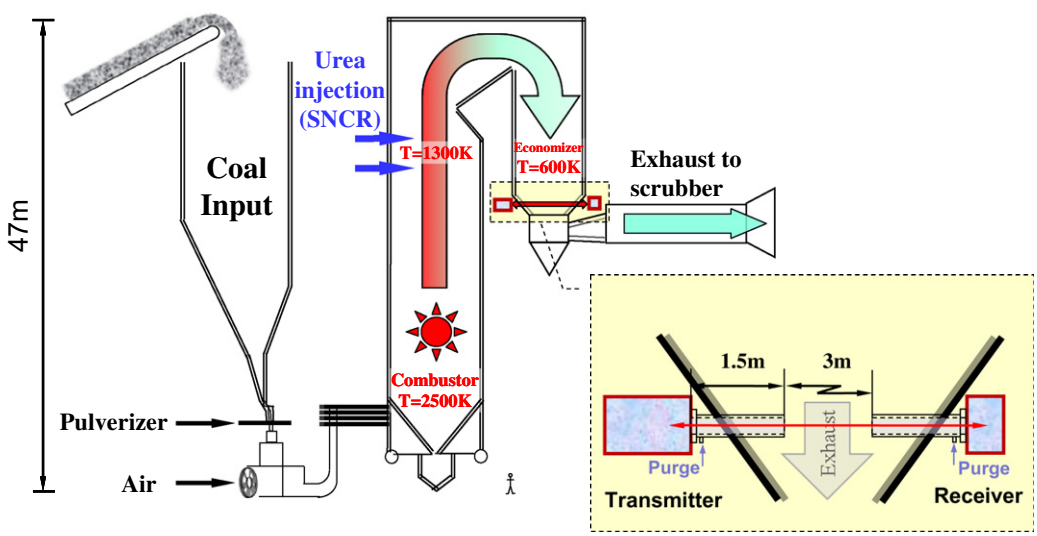


Fig. 46. Pulverized-coal fired boiler showing combustor, injectors for urea SNCR, bullnose, economizer, and location of diode laser absorption sensor for NO at the economizer exit where the gas temperature is ~600 K.

absorption sensing has been reported using transitions in all three domains: in the second overtone [195,196], the first overtone [197], and the fundamental band [198–200]. However, sensitive NO detection (ppm) using the overtone bands is not possible in combustion effluent due to the strong interference of hot water vapor even at the modest elevated temperatures (~600 K) of the economizer outlet.

The first NO detection by tunable-diode-laser absorption in flames using fundamental-band transitions was conducted over 25 years ago [198]; unfortunately, the only lasers near 5.2 μm at that time were low-power, multimode, cryogenically cooled lead-salt devices. More recently, NO detection in the mid-infrared has benefitted from the improvement of room-temperature quantum-cascade laser technology, now available both in pulsed and continuous-wave devices. McManus et al. evaluated field instruments for atmospheric air quality that included absorption detection of NO in air samples in a multipass cell at 1900 $cm^{-1}$ using both pulsed and CW QC lasers [199], and sub ppb-m$(Hz)^{-1/2}$ detection limits were reported. The QC laser technology was found reliable for long-term use. To illustrate the potential for mid-IR absorption sensor for sensing NO in combustion exhaust gases, Wysocki et al. [200] reported 1σ noise-equivalent sensitivity of 200 ppb-m$(Hz)^{-1/2}$ at 675*F* (630 K) with a variety of gas mixtures in a heated cell.

Recent experiments at Stanford to evaluate water vapor interference at elevated temperatures were conducted using a novel external-cavity quantum-cascade diode laser that could be tuned over the R-branch of the fundamental absorption band near 5.2 μm [194,200]. Laboratory measurements of NO were then performed in combustion exhaust gases where a NO detection limit of <60 ppb-m$(Hz)^{-1/2}$ was obtained at 600 K with a 1.79 m optical path. These laboratory measurements identified three promising NO doublets (R6.5, R10.5, and R15.5) for detection of ppm levels of NO in the presence of ~10% water vapor at temperatures ~600 K.

Measurements were then conducted using the Stanford sensor at the economizer exit of a pulverized coal-fired boiler as illustrated in Fig. 46 [194]. The measurement location was ~20 m above the floor of the combustor and the ambient temperature was ~30–50 °C. The mid-IR light from the QC laser was directed across the duct on a 3 m optical path. The boiler was equipped with urea injection for SNCR at the location noted in Fig. 46. Measurements over a one hour period are shown in Fig. 47 during which time the combustor excess air was increased from 2.5% to 3% and the urea injection was interrupted for 20 min. The measured NO followed the trends expected for these changes in firing conditions. The measured NO first increased with the addition of more excess air, then decreased again as the urea injection increased in response to increased NO and $O_2$ detected by the continuous emission monitoring near the exit of the stack. When the urea injection was stopped the measured NO rapidly increased and then when urea was reintroduced, the NO rapidly decreased, and the expected small overshoot of this decrease was captured. The total $NO_x$ measured by gas sampling at the exit of the stack is also shown in Fig. 47. The laser measurements agreed quantitatively with the sampled $NO_x$, but the real-time laser sensor avoided the ~5 min delay in the sample gas data due to the transit time of the exhaust gas from the economizer exit though the exhaust cleanup prior to the CEM sampling. This new 5.2 μm *in situ* sensor thus shows excellent promise for real-time monitoring of NO in combustion exhausts of large-scale coal-fired systems at temperatures up to at least 700 K.

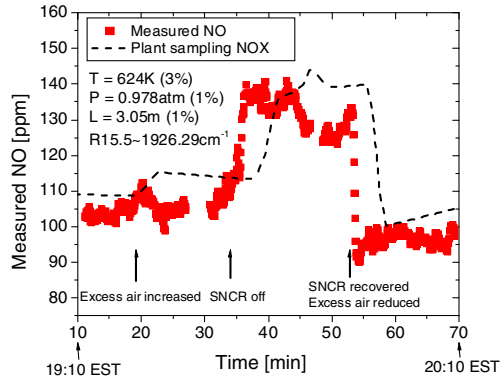


Fig. 47. Time-resolved NO measured at the economizer outlet by laser absorption and total $NO_x$ via sampling and chemiluminescence analysis near the final exhaust outlet (note the ~5 min delay) (figure adapted from Ref. [194]).

All of the applications discussed above have been research efforts that show proof-of-concept for laser-absorption sensing in large-scale combustion systems. However, sensors to monitor ammonia slip following $NO_x$ reduction using SNR or SNCR are already emerging in the commercial marketplace. The closed-loop control potential for SNR and SNCR reduction of $NO_x$ using $NH_3$ monitoring has been recognized for nearly two decades [171], following the $^{13}CO_2$ laser monitoring of $NH_3$ that was apparently the first application of laser absorption to coal-fired boiler monitoring [170,191]. In one application, diode laser absorption sensing of exhaust $NH_3$ from a 460 MW pulverized coal-fired boiler using SCR to reduce $NO_x$ found ammonia slip correlated with load, illustrating the potential for closed loop control [201]. Although diode-laser monitors of ammonia have been commercially

available for more than a decade, their application to large-scale coal-fired boilers have been slowed by a variety of challenges including variability of optical path lengths, strong scattering by flue-gas ash, and maintenance issues such as laser alignment, calibration and window fouling. Good sensitivity is needed for $NH_3$ slip sensing, as the industry desires to control the $NH_3$ below 2 ppm, and ideally a 0.1 ppm detection limit with 0.05 ppm precision is desired. Searching the web for "diode laser sensors for ammonia" finds more than ten vendors willing to offer commercial $NH_3$ sensors. All these instruments appear to use the near-infrared absorption band of $NH_3$ in the telecommunications band near 1.56 μm. However, a search of chemical abstracts for "$NH_3$ slip monitoring for SNR or SNCR" finds only two papers that report field measurement campaigns of *in situ* monitoring of $NH_3$ slip in the hot exhaust gases of coal-fired boilers [202,203]. The field testing reported for both sensors measured $NH_3$ via a cross-duct optical path near the economizer exit of a coal-fired boiler. In both cases the $NH_3$ detectivity was less than 1 ppm, the precision was better than 1 ppm, and in one case, the measurements compared well with gas sampling at the same location [203]. Both of these reports are more than five years old and both vendors advertise newer sensor versions than the ones reported. However, engineering issues remain, as none of the advertised devices appears to incorporate automatic beam pointing for closed-loop optimization of the optical alignment. Such technology has been used in the proof-of-concept demonstrations discussed above for gas- and coal-fired boilers and proved necessary for the large mechanical expansion that occurred in the transition from cold-shut-down to steady-state operations [191]. Therefore, one can anticipate that given the cost/benefit of lower maintenance, $NH_3$ laser sensors will ultimately be used routinely for commercial large-scale boilers, including automatic alignment systems.

### *4.5. Future of laser absorption sensors for large-scale combustors*

#### *4.5.1. Accomplishments*

The technology for diode laser absorption sensing is maturing rapidly. A 1998 review [204] focused on techniques for absorption sensing, the emerging diode laser sources, and applications in ambient air. During the past decade diode lasers have become available for nearly every wavelength from 0.6 to 10 μm, and a wealth of sensor research and development has been published. In this section of the present paper the discussion has focused on proof-of-concept measurements using sensors to control practical combustion devices. The earliest of this work attacked the high cost/benefit applications of specialty incinerators for shipboard and hazardous waste applications. The technology to monitor CO, $H_2O$ and gas temperature in these devices is now well-established, and sensors of CO or simultaneous $CO/O_2$ with or without gas temperature could be used in control systems needed to insure efficient incineration.

Sensors for larger-scale combustors require more engineering, and coal-fired applications are plagued by very low transmission over long paths due to scattering from particulate in the flow (e.g., unburned coal and ash in the combustor and flue-gas ash in the exhaust). However, the availability of increased laser power and use of larger collection apertures could efficiently mitigate these problems. As a result of the extensive research and development of laser absorption sensors over the past two decades, diode laser sensors to tune boiler combustors, monitor ignition safety, and optimize SNCR for $NO_x$ control are now emerging as commercial products.

#### *4.5.2. Needs and opportunities*

CO is a marker of incomplete or fuel-rich combustion and thus provides an indicator of burner malfunction or a control variable for air dampers (especially when simultaneous measurements of $O_2$ are available). As discussed above, a CO monitor was useful in controlling incineration efficiency, and measurements of CO were shown to correlate with slagging in coal-fired applications. However, monitoring of CO in the combustion zone offers a new opportunity for safe operation of flames with fuel streams of variable heating value. For safety reasons, nearly all large-scale combustors are fired with a few per cent of excess oxygen. However, the oxygen is usually monitored with solid-state sensors far downstream of the combustor. In the event of a burner malfunction or blowout the combustor can fill with explosive mixture faster than the response time of normal flame sensors. This can lead to an explosive result with huge costs in capital, operating revenue and potentially loss of life. Therefore, rapid-response, laser-based sensors of flame stability and combustion efficiency could be a very important development. These safety issues are exacerbated when the heating value of the fuel stream is uncertain, and many practical large-scale combustors suffer this problem, e.g., refineries, municipal waste incinerators, combustors operating on coal gasification products, some coal-fired boilers, or bio-fuel-fired combustors. For safe operation, these large-scale devices sacrifice efficiency by operating at higher excess air (and thus exhausting large volumes of excess hot air).

Laser-based monitoring of temperature fluctuations, with or without simultaneous CO and $O_2$ measurements in the combustion zone, offer a

potential solution to this problem. Except for coal-fired combustors, the CO concentration in the combustion zone is too low for detection in the telecommunications laser region due to the amount of water vapor interference. However, the recent availability of fiber-coupled diode lasers near 2.3 μm and new QC lasers near 4.4 μm offer potentially attractive solutions to sensitive monitoring of CO in the combustion zone.

The move toward lesser geopolitical dependence on oil resources in the US and China suggests that coal gasification products will become an important industrial fuel. Here the opportunity includes measurements at high pressures in the presence of large losses from particulate scattering. Recently, the first known application of diode-laser-absorption sensing was demonstrated in a biomass gasifier [205] Although, much research remains before these sensors are ready for closed-loop gasifier control, these first results offer promise for a new paradigm of gasifier control.

## 5. Conclusions

Quantitative, species-specific laser sensors have become essential tools in combustion research and development, and their use will become even more widespread in the future. Sensors based on molecular absorption and employing wavelength-tunable lasers are particularly attractive and will see increasing use owing to their relative simplicity and affordability as well as their compatibility with optical fibers for ready transmission of light to remote and distributed locations including hostile environments. The continued development of tunable diode lasers will lead to use of these light sources to access additional combustion species through absorption by rovibrational transitions in the infrared and eventually by electronic transitions in the ultraviolet [206]. Continued progress can also be expected in ability to monitor thermodynamic and gasdynamic properties such as temperature, pressure, and velocity as well as derived properties such as mass flux and momentum flux or thrust, of high importance to advanced propulsion systems. The capability of CW laser sources for continuous recording of species and properties will enable advances in diagnostic strategies that incorporate fluctuation analysis of real-time data, with potential implications for new forms of tomography to achieve spatial resolution. Certainly the use of wavelength multiplexing for simultaneous monitoring of multiple species and properties will grow, as the spectral range accessible with laser sources and optical fibers expands.

Availability of commercialized laser sensor systems will certainly improve, including dedicated and rugged systems for combustion monitoring that will function with minimal maintenance. Use of such systems for routine monitoring of combustion properties, such as power plant emissions, will expand, as will applications of such systems for combustion and propulsion control, extending in some cases to flight. Improved availability of commercialized laser-absorption sensors will allow use by individuals and groups with little training or interest in spectroscopy but with a primary focus on combustion or propulsion science and engineering.

New capabilities for combustion- and propulsion-related sensing will appear as researchers pioneer diagnostic techniques for monitoring new quantities, such as jet noise or real-time monitoring of thrust in pulsed propulsion systems, or develop new approaches for combustion control.

Lastly, advances in laser-induced fluorescence, as an adjunct to absorption, will also be made, certainly including continued development for high-speed imaging. And we may expect to see a class of laser sensor systems that combines the attributes of quantitative absorption with those of PLIF imaging.

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